

WORKS MANAGEMENT

for the construction of an

UNDERGROUND CBRN-HARDENED BLAST-RESISTANT PROTECTIVE STRUCTURE AND SENTRY POST

Pune, Maharashtra

Complete project — mobilisation to handover

Project	Underground CBRN-hardened, blast-resistant protective structure with associated sentry post
Location	Pune, Maharashtra
Objective	Protect 9 occupants for 96 hours against a nuclear air-blast design basis threat, with CBRN, EMP and fallout hardening
Design basis	Architectural Rev F · Structural Phase 2 Rev A + M1
Discipline packages	Drainage DR1 · HVAC HV1 · Schedule of Finishes FN1 · Structural CAD SC1
Works Management revision	WM1 , 07 September 2026
Programme	279 activities · 18 milestones · 404 logic links
Construction period	02 November 2026 to 20 November 2027 — 326 working days
Academic context	B.E. Civil Engineering, final semester

DESIGN CHANGE CARRIED BY THIS PACKAGE

The sentry post walls are **brick masonry**, replacing the 200 mm reinforced concrete ballistic infill panels of architectural revision F. This is the only design change carried by the Works Management package and it is recorded throughout as reference **SP-B1**. Every other element of the project follows the current design without alteration.

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This handout summarises a Works Management package whose detailed documents accompany it: the work breakdown structure, the bill of quantities and its full quantity derivation, the resource plan, the procurement plan, the quality plan, the safety and risk register, the codes register, the construction methodology, the progress-monitoring system, the assumptions and verification register, the master programme in Microsoft Project format and the programme drawing.

1 Introduction

This document sets out how the underground CBRN-hardened blast-resistant protective structure at Pune, together with its associated sentry post, is to be built and managed from site possession to handover. It is the Works Management component of the project and it covers the whole of it — the buried shelter, the entry structures, the escape shafts, the engineered cover, the sentry post, the services, the external works and the concealment works.

The design itself is not the subject of this document. The structure has been designed and detailed: a 22.0 × 6.2 m buried reinforced concrete box with 600 mm walls, a 900 mm pressure slab and a 600 mm mat, designed for a 344.7 kPa nuclear air-blast design basis threat under 2.0 m of layered engineered cover, together with an above-ground entry headhouse, a covered approach stairwell, two escape shafts and a separate two-storey sentry post. What this document addresses is the question that follows the design: how is it actually constructed, in what order, with what resources, to what quality, at what risk, and how is progress measured while it happens.

The starting point was a preliminary master construction schedule supplied with the project. That schedule contributed its project start date, its six-day working calendar and its practice of naming the structural consultant's pre-pour checks as distinct activities, and all three are retained. Its structure, however, described a different building: a 1000 mm roof slab under 4 m of overburden, a lift shaft, wall tiling and suspended ceilings, and no identifiable sentry post, escape shafts, blast doors, CBRN plant, EMP works or procurement. The programme has therefore been rebuilt around the project as it now stands rather than tidied up.

1.1 What makes this project unusual to build

Three characteristics drive almost every decision in this package.

- **It is a protective structure, so its continuity matters more than its strength.** There is no movement joint anywhere inside the protective envelope, because a movement joint is simultaneously a blast, a gas and an electromagnetic pulse discontinuity. Every construction joint carries two waterstops and a welded strap. Bar spacing is held at 150 mm in both curtains throughout — an electromagnetic pulse requirement, stricter than IS 456 requires for strength, and therefore one that a site engineer looking only at strength would see no reason to keep.
- **It is buried, so most of it can never be inspected again.** The external tanking is a continuous tank under the mat, up the walls and over the roof, and the design groundwater table sits at the roof level. Once it is backfilled the membrane is unreachable. The quality plan is built around that fact.
- **Several of its components cannot be bought quickly.** The blast doors, the CBRN filter trains, the blast valves and the shielded enclosure are specialist items with lead times measured in months, and the blast door cast-in frames have to be welded into the wall reinforcement before those walls are poured. Procurement is therefore not an administrative activity that runs alongside construction; it is part of the construction logic.

1.2 How this package handles what is not known

The project record classifies every value it holds as confirmed, reconstructed, assumed, unresolved or not available. This package uses the same classification and observes the same rule: nothing marked unresolved or not available is quietly filled in.

That rule has real consequences here. There is no electrical design package in this project, so every electrical quantity in the bill of quantities reads *to be verified from final measurement* and no electrical enquiry can be issued. There is no site plan, so the berm volume, the access route and several drainage runs cannot be measured. There is no lintel design for the sentry post openings — a requirement that the brick masonry itself creates. Each of these is recorded, dated against the programme activity it blocks, and issued to the designer at mobilisation rather than discovered when the activity is due to start.

WHAT THIS PACKAGE DOES NOT CLAIM

Two claims are never made about this structure, and are not made here.

A static analysis gives the demand on a blast structure, not proof of blast resistance. The non-linear support-rotation check is later-phase work and has not been done. This package delivers the structure that was designed; it does not validate the design.

Flotation cannot be read off a model whose springs take tension. It is a hand check and it belongs to the designer.

2 Project scope

The project comprises three physically separate structures with different design bases, different risk profiles and different positions in the programme. Treating them as one building is the mistake the preliminary schedule made and is the first thing this package corrects.

	Structure	Design basis	Position in the programme
1	Main underground box Buried RC shelter, eight bays, 22.0 × 6.2 m external, roof 2.0 m below grade, 20.8 × 5.0 m internal, 3.2 m clear height	Nuclear air-blast, p_{so} 344.7 kPa, design pressure 383 kPa on the roof <i>and</i> the walls. Total roof load 448.15 kPa. M35 concrete, Fe500D reinforcement	The controlling chain. WBS 3 and 4
2	Entry headhouse and covered approach stairwell Above-ground bermed RC on the shelter roof and adjacent fill	Headhouse walls and roof designed for full blast pressure on either face. The covered stairwell is outside the protective boundary and is declared expendable — designed to IS 456 with normal partial factors	Follows the pressure slab. WBS 5
3	Sentry post Separate two-storey RC framed building, 4.0 × 5.0 m, at least 10 m clear of the shelter excavation, on its own footings on in-situ basalt	Seismic Zone III, $R = 3.0$, $V_b = 73.18$ kN. M30 concrete, ductile detailing to IS 13920. Not blast designed — a recorded decision. Walls are brick masonry (SP-B1)	Independent, with 118 days of float. WBS 8

Table 2.1 — The three structures.

2.1 The protective boundary

The protective boundary is blast doors 1 and 2 at level (–)6.100, together with walls W6 and W7, the perimeter walls, the mat and the pressure slab. Everything above the blast doors — the stair shaft, the headhouse, the covered entry stairwell — lies outside it, and the covered stairwell is deliberately declared expendable.

That single definition produces a large part of the construction logic. It is why the blast door frames must be cast into W6 and W7 rather than fixed afterwards; why there is no movement joint inside the envelope but there is one where the stairwell raft meets the headhouse; why the gas-tight envelope tested at handover is bays 1 to 6 only, 67.8 m² of floor and 217.0 m³ of volume; and why the stair shaft and the generator bay sit outside that envelope as a deliberate grey zone.

2.2 Scope of the Works Management package

Scope element	Covered by	WBS
Main underground shelter structure	Mat, perimeter walls, protective walls W5/W6/W7, partitions, pressure slab with its openings, sump pit, main staircase	3, 4
Sentry post	Footings, frame, brick masonry walls , lintels, spiral stair, finishes, electrical	8
All associated structural works	Entry headhouse, covered entry stairwell, escape shaft collars	5
Access and egress	Main staircase, entry stairwell, blast doors, security door, external door, escape shafts	4.4, 5
External works	Berm, regrading, surface water, access, hardstanding, restoration	13
Drainage	Internal gullies and drains, clean sump and pumps, segregated decontamination effluent, septic tank, soak pits	10
Waterproofing	External tanking, roof membrane, protection screed, internal wet-area tanking, waterstops	6
Electrical works	Containment, cabling, distribution, lighting, earthing, EMP enclosure and penetration protection	11
Overburden and backfilling	Side backfill, six-layer engineered cover, M30 burster slab	7
Camouflage and concealment	Topsoil and turf, spoil dressing, screening, track discipline	13.3
Construction and completion	Mobilisation, earthworks, testing, commissioning, training, handover	1, 2, 14

Table 2.2 — Scope coverage.

3 Project components

Before a work breakdown structure can be written, the components that actually make up the finished project have to be identified from the project record rather than assumed. Twenty-three components are confirmed by the master project state file, the revision F drawings or an existing discipline package. All twenty-three are covered by this package.

Ref	Component	Source	Class
PC-01	Main underground box — mat foundation	master A.4.2, B.3, F.1	[C]
PC-02	Main underground box — perimeter walls W1–W4	master A.4.2, B.1, F.1	[C]
PC-03	Main underground box — protective walls W6 and W7	master A.3, B.2, F.1	[C]
PC-04	Main underground box — wall W5 and partitions W8	master A.3, F.1	[C]
PC-05	Main underground box — pressure (roof) slab	master A.4.2, A.7.4, B.4, F.1	[C]
PC-06	Roof openings — stair void and escape shafts	master A.4.4, A.4.5, B.4.1, F.1	[C]
PC-07	Sump pit SU-01	master F.1, Drainage DR1	[C]/[U]
PC-08	Escape shafts ESC 1 and ESC 2	master A.4.5	[C]
PC-09	Blast doors 1 and 2	master A.2, F.1	[C]/[N]
PC-10	Main staircase — FROZEN	master A.4.4, B.5; frozen by project instruction	[C]
PC-11	Entry headhouse	master A.4.6, B.7, F.3	[C]
PC-12	Covered entry stairwell	master A.4.7, B.6, F.2	[C]
PC-13	Engineered cover / overburden	master A.7.3	[C]
PC-14	Waterproofing system WP-01 to WP-06	Schedule of Finishes FN1, R-805	[C]
PC-15	Drainage and sanitary installation	Drainage DR1	[C]
PC-16	HVAC and CBRN plant	HVAC HV1	[C]
PC-17	Electrical and EMP installation	master A.3, G; MIL-STD-188-125-1	[C] scope / [N] design
PC-18	Internal finishes	Schedule of Finishes FN1	[C] require ment / [N] product
PC-19	SENTRY POST — RC frame	master A.4.8, B.8, F.4	[C]
PC-20	SENTRY POST — BRICK MASONRY WALLS (design change SP-B1)	SP-B1, this package; geometry from Rev F drawings 3, 4, 6	[C] instruction / [D] quantities
PC-21	External works, berm and site restoration	master A.4.6, A.4.7; Drainage DR1 open item D3	[C] intent / [N] extent
PC-22	Camouflage and concealment	master A.7.3	[C] layer / [N] regime
PC-23	Testing, commissioning and handover	master G; this package	[C]

Table 3.1 — Confirmed project components. Full descriptions are in the project component register.

3.1 Components that do not exist

A component is included only where the project record supports it. The preliminary schedule contained several that it does not, and each was removed rather than carried forward.

Item in the preliminary schedule	Why it is not in this project
Lift / lift shaft	The supplied R0 schedule contains 'Column, Staircase & Lift Shear Wall' activities. THERE IS NO LIFT ANYWHERE IN THIS PROJECT. Those activities are generic building boilerplate and have been removed.
Second grade slab over 4 m of overburden	The R0 schedule programmes a '2nd Slab (Ground Lvl) Grade Slab' over '4m Soil Overburden'. The current design has a SINGLE 900 pressure slab at (-)2.000 under 2000 of layered engineered cover containing a 200 M30 burster slab. The 4 m arrangement is a superseded revision.
1000 mm roof slab	The R0 schedule calls the roof '1st Slab (1000 mm Thk)'. The confirmed thickness is 900.
Structural RC columns in the shelter	None exists. The box is a wall-and-slab structure; master B.3 declares punching shear NOT APPLICABLE for the same reason, and Structural CAD SC1 records that drawing a column sheet would be fabrication.

Item in the preliminary schedule	Why it is not in this project
Dado / wall tiling in the shelter	The R0 schedule programmes wall tiles. The Schedule of Finishes states the opposite for the gas-tight envelope: 'NO joints, NO tiles, NO grout — grout lines are a decontamination failure'. Tiling is therefore not programmed inside the envelope.
Suspended ceilings and gypsum punning in the shelter	The R0 schedule programmes gypsum punning to ceilings. FN1 forbids any suspended ceiling or boxing-in in the gas-tight envelope for three confirmed reasons: duct inspectability, pipe inspectability and contamination trapping.

Table 3.2 — Items removed from the preliminary schedule.

The preliminary schedule also contained no sentry post as an identifiable structure, no escape shafts, no blast doors, no CBRN plant, no electromagnetic pulse works, no engineered cover build-up, no burster slab, no concealment works, no procurement activities, no commissioning sequence, and a single milestone. All of that is project scope and all of it is now in the breakdown.

4 Construction methodology

The methodology is set out in full in the accompanying construction methodology document. This section gives the decisions that shape it — the ones that produce the logic links in the programme.

M1 Site establishment and setting out

The site is a military installation, so the boundary, gate and access control are established before any other activity and workforce security clearance runs in parallel with mobilisation.

Setting out uses the project's own coordinate system (master A.4.1): origin at the south-west external corner of the box at ground level, X east 0 to 22000, Y north 0 to 6200, levels in metres relative to finished grade 0.000. Two permanent benchmarks are established outside the excavation influence zone and checked independently.

M2 Excavation

Topsoil is stripped 150 deep and STOCKPILED, not carted away: the same material returns as the 300 concealment layer at the top of the engineered cover.

Excavation is 6.800 m deep over a 24.000 x 8.200 envelope, which includes 1.000 m of working space outside the external wall face. That working space is not optional — the external tanking membrane and its protection are applied from outside the structure.

Only the top 1.5 to 2.0 m is soil; below rockhead the material is Deccan basalt. Rock is removed by HYDRAULIC BREAKER, not by blasting. The sentry post founds on the same rock within 10 m, the site is operational, and controlled blasting would require a vibration regime the project has not specified.

M3 Dewatering

The design groundwater table is (-)2.000 in the monsoon and is ASSUMED — the master calls it the single most important number the site investigation must confirm. Of the 15.41 kPa/m lateral gradient on the walls, 9.81 is water and only 5.60 is soil.

Construction dewatering runs continuously from the start of the deep excavation until the side backfill is complete, with a standby pump and an independent power source. Groundwater monitoring during the monsoon is programmed as a critical-path activity precisely because the answer changes the dewatering, the tanking and the flotation check.

Discharge is to the surface water system away from the excavation, through a settlement facility. Silt-laden discharge into the soak pits would destroy their percolation capacity before they are commissioned.

M4 Blinding, tanking and the mat

100 M15 blinding is placed over the whole formation, projecting 100 beyond the mat, to give a clean surface for the membrane.

The external tanking WP-01 is a CONTINUOUS TANK: horizontal on the blinding, turned up the external wall faces, and lapped to the roof membrane. It is inspected and signed off before it is covered, because after that it can never be seen again.

The mat is cast in a single continuous pour of about 85 m³ including the sump pit, which is cast monolithic with it and has the membrane dressed around it.

M5 Walls

The 3.200 m wall height is cast in two lifts of 1.600 m. Vertical construction joints are formed at about 6 m centres; at every joint the reinforcement is FULLY CONTINUOUS and two waterstops plus a welded copper or galvanised EMP strap are installed. A construction joint in a protective structure is simultaneously a water path, a gas path and an EMP discontinuity, and all three have to be closed.

There is no movement joint anywhere inside the protective envelope. The only movement joint on the project is where the covered stairwell raft meets the headhouse — outside the boundary, in the structure that is declared expendable.

Bar spacing is 150 both curtains throughout. This is an EMP requirement and is stricter than IS 456 — it is not a strength requirement and must not be relaxed on site to ease congestion.

M6 Pressure slab

The pressure slab is the governing element: 900 thk, T25 @ 150 each face each way, 20.5 t of reinforcement, carrying 448.15 kPa. It spans ONE WAY across the 5.000 internal width, which is the single fact that makes lengthening the box cheap and widening it expensive.

Falsework carries 104 m² of deck at 3.200 m height and is designed and checked as temporary works.

Reinforcement is fixed in a fixed order: bottom curtain, opening trimmers, escape shaft collar bands, local thickenings, links, top curtain, haunches. The opening details are not incidental — the stair void carries 6-T25 top and bottom with 4-T25 diagonal bars at each re-entrant corner, and each shaft carries 5-T25 on every side, face and direction with the slab thickened 900 to 1200 over a 600 annulus.

M7 Entry structures and escape shafts

The headhouse is built off the top of the pressure slab, its floor being the slab surface at (-)2.000. Wall HW4 aligns exactly with W7 after modification M1; HW3 has no wall beneath it and loads the slab as a line load, which the slab was checked for.

The headhouse roof is 500 thk at 90 % utilisation — the most highly worked element in the project. Its reinforcement inspection is a hold point attended by the structural consultant.

The covered entry stairwell is built on a stepped raft on compacted fill, with the movement joint at the headhouse. It is outside the protective boundary and is DECLARED EXPENDABLE; it is built to IS 456 with normal partial factors and is not blast rated.

M8 Backfill, engineered cover and overburden

The walls are NOT backfilled until the pressure slab is cast and cured. A 3.2 m wall carrying an 83.2 kPa gradient at its base needs the roof as its prop; backfilling earlier means designing and installing temporary propping to do the same job worse.

Side backfill is placed in 250 layers compacted to 95 % MDD and tested in situ before the cover goes on.

The engineered cover is a designed six-layer structure, not a filling operation, and it is built from the bottom up in the reverse of the order given in master A.7.3: 100 protection screed over the membrane, 750 compacted engineered fill at 95 % MDD (the radiation mass), 500 crushed basalt rubble at 25–75 mm (which scatters burster energy), the 200 M30 burster slab reinforced T12 @ 150 both ways (which breaks up a penetrating item), 150 granular filter (which stops fines clogging the rubble) and 300 topsoil and turf (concealment and erosion control).

M9 Sentry post — frame

The sentry post is an independent structure and is deliberately NOT started until the main rock breaking is finished. No green concrete stands within 10 m of a hydraulic breaker.

Footings F1 found on in-situ basalt at (-)2.000; the founding stratum is a hold point in the same way as the main formation.

Columns, beams and slabs are detailed to IS 13920 for Zone III. The detail that most often gets lost on site is the column confinement: T10 hoops with a cross-tie each way at 85 c/c over 500 mm from EVERY joint face, top and bottom, AND THROUGH THE JOINT.

M11 Services

Builder's work — sleeves, ducts and penetrations — is coordinated with the reinforcement and is a predecessor to the wall inspection hold point. Cutting a penetration afterwards through a 600 wall at 150 bar spacing both curtains is not a repair, it is a defect.

Every duct and every pipe in the gas-tight envelope must remain inspectable along its whole length. That is why there is no suspended ceiling and no boxing-in anywhere in bays 1 to 6.

The decontamination effluent system is physically segregated from the clean sump and is never connected to it. Effluent goes to tank TK-01 for tanker removal.

M12 Finishes

Every finish in this project is specified as a PERFORMANCE REQUIREMENT and the product is deliberately left open. The requirements come from the decontamination duty, the gas-tight envelope, the EMP requirement and the wet areas.

Floors are laid to falls, joint-free within a room, and coved into the skirting in one piece. In the wet areas — bays 2, 5 and 6 — the finish is fully impervious and coved 150 up every wall in a single piece, with no tiles and no grout, because a grout line is a decontamination failure.

M13 External works and concealment

The berm is graded 1.5:1 to +0.900 against the headhouse and covered stairwell, and the site is crowned with 1:50 falls away from the structure.

Turf over the cover is the confirmed concealment measure. Anything beyond it — spoil dressing, screening, planting regime, track discipline — is programmed as work but has no specification in the project, and the extent must be confirmed before it is priced.

M14 Testing, commissioning and handover

Structural verification is by 28-day cube results supported by ultrasonic pulse velocity and rebound hammer testing.

The protective systems are tested in a fixed order because each depends on the one before: EMP shielding effectiveness survey, blast door functional and seal test, then the gas-tightness and overpressure test of bays 1 to 6, then CBRN filter train commissioning, then integrated protective-mode operation.

5 Work breakdown structure

The breakdown has three levels: fourteen level-1 work packages, 60 level-2 sub-packages and 279 level-3 activities of which 18 are milestones. It runs from contract award to practical completion and it covers all three structures in one integrated programme rather than three separate ones.

WBS	Level-1 work package	Acts	Start	Finish	Days
1	PRE-CONSTRUCTION AND ENABLING WORKS	37	02-11-26	04-05-27	155
2	SITE PREPARATION AND EARTHWORKS	17	26-11-26	11-05-27	140
3	MAIN UNDERGROUND SHELTER — SUBSTRUCTURE	35	06-02-27	25-05-27	92
4	MAIN UNDERGROUND SHELTER — PRESSURE SLAB	22	05-05-27	26-08-27	98
5	ENTRY STRUCTURES, ESCAPE SHAFTS AND DOORS	29	28-06-27	15-09-27	69
6	WATERPROOFING	9	05-05-27	17-09-27	117
7	BACKFILL, ENGINEERED COVER AND OVERBURDEN	12	21-01-27	10-09-27	198
8	SENTRY POST	42	21-01-27	11-06-27	120
9	MECHANICAL, CBRN AND HVAC SERVICES	11	20-03-27	26-08-27	136
10	DRAINAGE AND SANITARY WORKS	12	23-07-27	16-09-27	48
11	ELECTRICAL AND EMP WORKS	10	15-03-27	02-09-27	147
12	INTERNAL FINISHES	13	27-07-27	18-09-27	47
13	EXTERNAL WORKS, CONCEALMENT AND RESTORATION	9	10-09-27	18-10-27	32
14	TESTING, COMMISSIONING AND HANDOVER	21	05-08-27	20-11-27	92

Table 5.1 — Level-1 work packages.

5.1 Coding rules

- **Level 1** is a work package that could be let separately or reported on separately to the client.
- **Level 2** groups activities sharing a work face, a trade and a set of hold points.
- **Level 3** is an activity one gang can be instructed to do, with a measurable quantity behind it, that can be marked complete without ambiguity.
- Milestones carry zero duration and no resource. They are reporting events, not work.
- Hold points are activities in their own right — they consume time and require the Engineer's attendance — and are named so that they cannot be passed over silently.
- Activity numbers follow the level-1 package: A1xxx pre-construction through to A14xxx testing and handover.

The breakdown and the programme are the same data. Every activity code in the work breakdown structure appears in the Microsoft Project file with the same identifier, duration and logic, and every item in the bill of quantities is measured against one of these work packages. They are generated together and cannot drift apart.

6 Underground shelter works

The shelter accounts for 124 of the 279 activities and for the whole of the critical path between formation approval and the completion of the engineered cover.

6.1 Principal quantities

Work	Quantity	Note
Bulk excavation, 0.000 to (–)6.800	1 338 m ³	24.0 × 8.2 m envelope including 1.0 m working space; vertical faces
— of which in soil / weathered overburden	344 m ³	at the mean rockhead (–)1.750; range 295–394 m ³
— of which in rock (Deccan basalt)	994 m ³	hydraulic breaker, no blasting; range 945–1 043 m ³
Blinding, M15	17.3 m ³	100 thk under the mat, footings and raft
Structural concrete, M35	388.7 m ³	mat 81.8 · walls 119.7 · pressure slab 112.0 · headhouse 32.7 · stairs and stairwell 22.9 · sump, collars, thickenings, partitions 19.5
Reinforcement, Fe500D	70.46 t	92 bar marks; 33.6 % T16, 28.8 % T12, 21.9 % T25, 15.5 % T20
Formwork, contact area	1 058 m ²	including 104 m ² of falsework deck at 3.200 m height
Waterproofing	532 m ²	continuous external tank plus internal wet-area tanking
Side backfill	290 m ³	250 layers to 95 % maximum dry density
Engineered cover	206 m ³	six layers totalling 2 000 mm

Table 6.1 — Principal shelter quantities.

6.2 The three programme drivers

WHAT CONTROLS THE SHELTER PROGRAMME

Rock excavation — 994 m³ removed by hydraulic breaker. No blasting is used: the sentry post founds on the same rock within 10 m and the installation is operational. Two breaker rigs at about 30 m³ a day give 20 working days across two lifts.

The pressure slab prop period — 14 days. IS 456 Table 11 requires 14 days for props to a slab spanning over 4.5 m. This slab spans 5.000 m clear. With 14 days of curing before it, 19 working days sit on the critical path and are not available for compression. Striking requires a signed permit for exactly that reason.

The engineered cover — a structure, not a fill operation. Six layers, one of which is a 200 mm M30 reinforced burster slab that has to be formed, reinforced, poured and cured inside the cover. Thirty-four working days from the roof membrane to the finished turf.

6.3 The main staircase

The main staircase geometry is fixed and is not touched by this package: 24 risers at 170.8333 mm, 280 mm tread, three flights of eight, total rise 4 100 mm, flights 1 200 mm wide, a 200 mm well and 2 533 mm headroom. It is inside the protective envelope but is not a blast element and is designed to IS 456 with normal partial factors.

It is programmed in two parts for a construction reason. The lower two flights and landings L1 and L2 are built immediately after the walls, which gives permanent access into the excavation and removes a temporary stair from the temporary works. The third flight rises to (–)2.000, which is the top of the pressure slab, so it cannot be cast until the slab is in and its props are struck.

7 Sentry post works

The sentry post is a separate two-storey reinforced concrete framed building 4.0×5.0 m on plan, at least 10 m clear of the shelter excavation, on its own footings on in-situ basalt at $(-)2.000$. It is not blast designed — that is a recorded decision in the project, not an omission — and it is designed for seismic Zone III with ductile detailing.

Element	Size	Reinforcement	Quantity
Footings F1	$1500 \times 1500 \times 600$, 4 No., on in-situ basalt at $(-)2.000$	T12 @ 150 both ways bottom	5.40 m ³ M30
Columns C1	350×350 , 4 No., both storeys, $(-)1.400$ to $+6.700$	8-T16; T10 hoops with a cross-tie each way @ 85 over 500 from every joint face and through the joint; T8 @ 150 elsewhere	3.97 m ³
Plinth beam PB	250×400 at $+0.450$, 15.20 m clear	3-T12 top + 3-T12 bottom	1.52 m ³
Beams B1	250×450 , spanning 3 650 c/c, grids 1 and 2	4-T16 top at supports, 2-T16 bottom continuous	included below
Beams B2	250×450 , spanning 4 650 c/c, grids A and B	3-T20 top at supports, 2-T20 bottom continuous — <i>3-T20 chosen over 5-T16 because five T16 bars need 256 mm in a 250 mm beam</i>	2.28 m ³ (ribs)
Slab S1	150 two-way, first floor 4.0×5.0 and roof 4.6×5.6 with the 300 projection	T8 @ 150 both ways bottom; T8 @ 300 edge top 400 into the span; T8 @ 200 torsion in four layers over 700×700 at all four corners	6.86 m ³
Parapet	300 high over the roof projection	nominal — not designed	0.92 m ³
Walls (SP-B1)	190 brick masonry in a 200 structural zone, both storeys	Unreinforced infill; lintels over all openings	12.20 m³
Spiral stair	External, 1000 R with a 250 dia central pole	Fabrication detail not in the project record	1 No.

Table 7.1 — Sentry post elements and quantities.

7.1 Position in the programme

The sentry post is deliberately not started until the main rock breaking is finished. No green concrete stands within 10 m of a hydraulic breaker, and holding the post back also shortens the period over which a finished building is exposed to main-works traffic.

It carries 118 working days of float — by a wide margin the largest block of float in the programme. That makes it the natural place to absorb labour redeployment when the main works are held up, and the natural place to take labour from when the main works need to catch up. It is reported as a separate progress stream for the same reason.

Milestone	Baseline date	Float
Sentry post frame complete	20 April 2027	182 d
Sentry post brick masonry complete	28 April 2027	175 d
Sentry post complete	11 June 2027	109 d

Table 7.2 — Sentry post milestones.

7.2 Reinforcement quantity

The sentry post has never been quantified anywhere in the project — the reinforcement package explicitly excludes it. Its reinforcement has therefore been derived here from the confirmed arrangement in the master reinforcement register: **1.97 t**, being 0.64 t of T8, 0.25 t of T10, 0.19 t of T12, 0.62 t of T16 and 0.27 t of T20. Against 20.95 m³ of concrete that is about 94 kg/m³, which is plausible for a small ductile-detailed frame whose columns carry T10 confining hoops at 85 mm centres. It is a sanity check, not a design check, and it is an estimating quantity for tender that must be replaced by the fabricator's approved bar bending schedule.

8 Sentry post brick masonry methodology

DESIGN CHANGE SP-B1

The sentry post walls are BRICK MASONRY. This replaces the '200 RC BALLISTIC INFILL PANELS' shown on Rev F drawings 3, 4 and 6 and described in master A.4.8. It is the only design change carried by this Works Management package and it is recorded as reference SP-B1.

Two consequences follow and are recorded rather than buried. First, brick masonry does not provide the ballistic protection that the Rev F panels were named for; the change removes a stated protective function from the sentry post, which is already a structure the project declares NOT blast designed and expendable. Second, the openings now need lintels, which the RC panels did not, and no lintel design exists anywhere in the project. Both are carried into the verification register for the designer to close.

8.1 Geometry

Item	Value	Basis
Wall run per storey	15.200 m	$4.000 - 2 \times 0.350 = 3.300$ on the north and south walls, $5.000 - 2 \times 0.350 = 4.300$ on the east and west walls, two of each
Panel height	2.600 m	the project's own confirmed figure — master A.7.7 gives the infill load as $13.000 \text{ kN/m} = 0.200 \times 2.600 \times 25$, verified
Structural zone	200 mm	unchanged from Rev F; the columns' outer faces are flush with the wall
Brickwork thickness	190 mm	one brick thick in modular bricks to IS 1077 ($190 \times 90 \times 90$), built inside the 200 zone with the residual 10 mm taken up at the internal face in the plaster, so the confirmed 4000×5000 external envelope and the column faces are preserved exactly
Ground storey openings	D1 900 wide; W1 1200 wide	D1 in the west wall at Y 1000–1900; W1 in the east wall at Y 1900–3100 — both read from the Rev F drawing
First storey openings	8 vision panels 1200 wide; D1 900	two panels per face on all four faces; the door lies within the west panel on the drawing, and the overlap is a verification item
Quantity	12.20 m³ / 64.19 m² face	6.88 m ³ ground storey, 5.32 m ³ first storey, net of openings
Materials	~6 400 bricks; 15 bags cement; 3.0 m³ sand	modular bricks class 10 to IS 1077 with 5 % breakage; CM 1:6

Table 8.1 — Brick masonry geometry and quantity.

8.2 Why 190 mm and not 230 mm

The confirmed structural zone between the column faces is 200 mm: the sentry post is $4\,000 \times 5\,000$ externally and $3\,600 \times 4\,600$ internally, with 350×350 columns whose outer faces are flush with the wall. Two standard Indian brick sizes are available. Modular bricks to IS 1077 at $190 \times 90 \times 90$ build a 190 mm one-brick wall that fits inside the 200 mm zone with 10 mm to spare, which is taken up at the internal face and absorbed in the plaster. Conventional bricks at $230 \times 110 \times 75$ build a 230 mm wall that would project 30 mm beyond the column faces and change the confirmed external envelope.

The modular brick is adopted because it preserves every confirmed dimension. Using the conventional brick would be a second design change, and the instruction for this package is that there is only one. The brickwork is measured at its actual 190 mm thickness, as IS 1200 (Part 3) requires, not at the 200 mm zone width.

8.3 Why the panel height is 2.600 m

This is the project's own confirmed figure, not an assumption. The master gives the infill load on the first-floor beams as 13.000 kN/m and verifies it as $0.200 \times 2.600 \times 25$, and the framing plan repeats '200 × 2600 high'. The first storey checks geometrically: the roof slab top at +6.700 less the 450 mm beam depth gives a soffit at +6.250, and $+6.250 - +3.650 = 2.600$. The ground storey does not: the first floor at +3.650 less 450 mm gives a soffit at +3.200, and $+3.200 - +0.450 = 2.750$, which is 150 mm more. The project's confirmed 2.600 is used for both storeys and the difference is carried as a verification item. It is declared, not resolved.

8.4 Construction method

1. Source Approval And Testing. Bricks are procured against IS 1077 as class designation 10 (10 N/mm² minimum average compressive strength). Samples are drawn to IS 5454 and tested to IS 3495 for compressive strength, water absorption, efflorescence and warpage before any brick is laid. On a military structure the source is approved once and not changed without retesting.

2. Stacking And Wetting. Bricks are stacked to IS 4082 clear of the working area and are thoroughly soaked before laying so that they do not draw water out of the mortar. Dry bricks in a CM 1:6 bed are the most common cause of weak masonry on Indian sites.

3. Mortar. Cement mortar 1:6 by volume for the general walling, prepared and used to IS 2250, with sand to IS 2116. Mortar is used within 30 minutes of mixing and is never re-tempered. Gauge boxes are used — proportioning by shovel is not accepted.

4. Bond And Coursing. English bond, one brick thick. Joints are 10 mm and are fully filled — bed, cross and vertical. Every fourth course is checked for level and every panel is checked for plumb. Courses are kept level and no part of the wall is raised more than one metre above another in a day's work.

5. Interface With The Frame. The masonry is an INFILL between RC columns and under RC beams. It is not a load-bearing wall and must not be built tight to the beam soffit in one operation: the top course is left down and is wedged and packed after the panel below has taken its initial shrinkage, so that the panel does not attract load from the frame it was not designed to carry.

6. Ties To The Columns. The masonry is tied to the columns so that the panel is restrained out of plane. The tie detail does not exist in the project and is carried as a verification item; until it is designed, the assumption for planning is starter bars or proprietary ties at every fourth course.

7. Seismic Note — Not Resolved Here. Master A.7.8 records that the sentry post uses $R = 3.0$ precisely BECAUSE the infill is not separated from the frame, and B.8 designs to $V_b = 73.18$ kN. Substituting brick masonry at about 20 kN/m³ for 200 RC at 25 kN/m³ reduces the infill line load from 13.000 kN/m to roughly 9.9 kN/m and therefore reduces the seismic weight, which points the existing design in the conservative direction. That is a direction, not a verification. The structural discipline must re-run the check; this package does not.

8. Lintels. RC lintels are required over D1, W1 and the eight vision panels. Bearing is taken as 200 each end for planning. NO LINTEL DESIGN EXISTS IN THE PROJECT — section, reinforcement and bearing are to be designed before the openings are built, and the 200 x 150 section used for the estimate is for quantity only.

9. Curing. Masonry is kept damp for at least seven days. Chases for electrical conduit are CUT, not hammered, and never cut into a course until the masonry has cured.

10. Tolerances And Inspection. Line, level, plumb and joint thickness are inspected at a hold point at the end of each storey before plaster. Plumb over the 2.600 storey height and joint thickness are the two measurements that decide whether the plaster can be held to its stated thickness.

8.5 Programme

Act	Description	Days	Start	Finish
A1176	Modular bricks to IS 1077 — source approval, testing and order	20	19-11-26	11-12-26
A8130	Brick delivery, stacking, sampling and testing to IS 3495	4	15-03-27	18-03-27
A8135	BRICK MASONRY WALLS — ground storey, 190 thk in CM 1:6	5	23-03-27	27-03-27
A8140	RC lintels over the D1 and W1 openings — ground storey	2	29-03-27	30-03-27
A8145	HOLD POINT: masonry line, level, plumb and joint thickness — ground storey	1	31-03-27	31-03-27
A8150	BRICK MASONRY WALLS — first storey, 190 thk in CM 1:6	4	20-04-27	23-04-27
A8155	RC lintels over the vision panel and door openings — first storey	2	24-04-27	26-04-27
A8160	HOLD POINT: masonry inspection — first storey	1	27-04-27	27-04-27
A8165	MILESTONE: sentry post brick masonry complete	—	28-04-27	28-04-27
A8175	Sentry post electrical — conduits and boxes chased into the masonry	4	28-04-27	03-05-27
A8180	Sentry post internal plaster 12 mm, sand faced, CM 1:4	6	04-05-27	10-05-27
A8185	Sentry post external plaster 15 mm, two coat, with scaffolding	6	11-05-27	17-05-27

Table 8.2 — The brick masonry sequence. Each storey is built only after the slab above it has cured, so the masonry is built into a completed frame rather than the frame being completed around it.

8.6 Verification items raised by this change

Ref	Item	Status
WM-V1	Ground-storey panel height: the project's confirmed 2.600 m against the 2.750 m implied by the levels	Open — 2.600 used
WM-V2	First-floor west wall: door D1 lies inside the vision panel on the drawing	Open — the union is deducted once
WM-V3	Window and vision-panel heights are not stated anywhere	Open — 1200 assumed
WM-V4	190 mm modular brickwork inside the 200 mm zone	Open — needs the designer's confirmation
WM-V5	Lintel design over the nine openings — a new requirement created by SP-B1	Open — structural design required
WM-V6	Seismic weight changes. Brick at about 20 kN/m ³ over 0.190 × 2.600 gives roughly 9.9 kN/m against the confirmed 13.000 kN/m, so the existing base shear is conservative — a direction, not a verification	Open — referred to the structural discipline
WM-V7	Ballistic performance. The panels are described as '200 RC ballistic infill'; brick masonry does not provide equivalent protection	Open — recorded so the consequence is visible
—	Wall tie detail between the masonry and the columns	Does not exist in the project

Table 8.3 — Verification items. None is resolved here.

9 Construction sequence

The sequence below is the programme logic in narrative form. The features worth noting are that the roof goes on before the walls are backfilled, that the sentry post is held back until the rock breaking is over, and that the protective systems are tested in a fixed order because each depends on the one before it.

Stage	Sequence
Mobilisation	Approvals and workforce security clearance; site boundary and access control first, because the site is operational. Survey and confirmatory site investigation, with monsoon groundwater monitoring deliberately on the critical path. Blast doors ordered on day one.
Earthworks	Clear and strip, keeping the topsoil for the concealment layer. Temporary works and surface water cut-off. Excavate the soil to rockhead, then break rock in two lifts to (–)6.800. Sump pit. Dewater continuously. Trim the formation, over-excavate any red-hole seam and replace with M15. Formation hold point.
Substructure	Blinding, then the horizontal tanking, then a hold point before it is covered. Mat and sump cast in one pour with the wall starters set. Fourteen days curing. Wall reinforcement in both curtains at 150 spacing, with the blast door frames set, aligned and welded to the cage. Walls in two lifts of 1.6 m with two waterstops and a welded strap at every joint. Lower stair flights, which give permanent access.
Pressure slab	Falsework at 3.2 m. Bottom curtain, opening trimmers, escape shaft collar bands, local thickenings, links, top curtain, haunches, cast-ins. Pre-pour hold point. A single continuous 112 m ³ pour. Fourteen days curing, then props for fourteen days to IS 456 Table 11. Strike, then the upper stair flight.
Entry structures	Headhouse walls and roof off the top of the pressure slab; the covered stairwell on its stepped raft with the movement joint at the headhouse; escape shaft collars up to their heads before the cover is placed; blast doors installed.
Waterproofing	Vertical tanking and its protection, with a hold point before backfilling because it can never be seen again. Roof membrane and the 100 mm protection screed, with a second hold point before covering.
Sentry post	Held until the rock breaking is over. Footings on a confirmed founding stratum, then the frame storey by storey. Brick masonry ground storey after the first floor has cured, first storey after the roof has cured, each with a hold point on line, level, plumb and joint thickness. Lintels, spiral stair, chases cut for conduit, plaster, flooring, finishes.
Services	Builder's work coordinated with the reinforcement, not cut afterwards. Ductwork with every run inspectable. Blast valves. Filter trains and generator. Drainage with the decontamination effluent physically segregated. Electromagnetic pulse enclosure and penetration protection. Electrical distribution and earthing.
Cover and overburden	Side backfill in 250 layers, density tested before the cover goes on. Then the six layers from the bottom up: 100 screed, 750 compacted fill, 500 crushed basalt won from our own excavation, the 200 M30 burster slab, 150 granular filter, 300 topsoil and turf. Verified against the design build-up.
External works	Berm at 1.5:1 to +0.900, site regraded with 1:50 falls, surface water, access and hardstanding, turfing, concealment measures, removal of temporary works.
Testing and handover	Cubes and non-destructive testing; watertightness; drainage; electrical and a 5 Ω earth; then the protective sequence in order — shielding survey, blast door seals, gas-tightness and overpressure, filter train commissioning, generator, integrated protective-mode operation. Snagging, rectification, contingency, documentation, training, handover.

Table 9.1 — Construction sequence.

9.1 Three sequencing decisions worth defending

The roof before the backfill. A 3.2 m wall carrying an 83.2 kPa gradient at its base needs a prop at the top. The pressure slab is that prop. Backfilling earlier would mean designing, installing and later removing temporary propping to do the same job worse, and would put temporary works loads into a structure whose whole purpose is that it carries a very large permanent one.

The lower stair flights early. Building the first two flights and landings immediately after the walls costs nothing on the critical path and removes a temporary access stair from a 6.8 m excavation for the whole of the pressure slab operation.

The sentry post after the rock breaking. Hydraulic breaking within 10 m of green concrete is avoidable simply by ordering the work differently, and doing so also reduces the time a finished building stands exposed to main-works traffic.

10 Final work programme

The master programme covers the complete project in one integrated network: 279 activities, 18 milestones and 404 logic links using finish-to-start, start-to-start and finish-to-finish relationships with positive and negative lags. It runs from 02 November 2026 to 20 November 2027 — 326 working days on a six-day week.

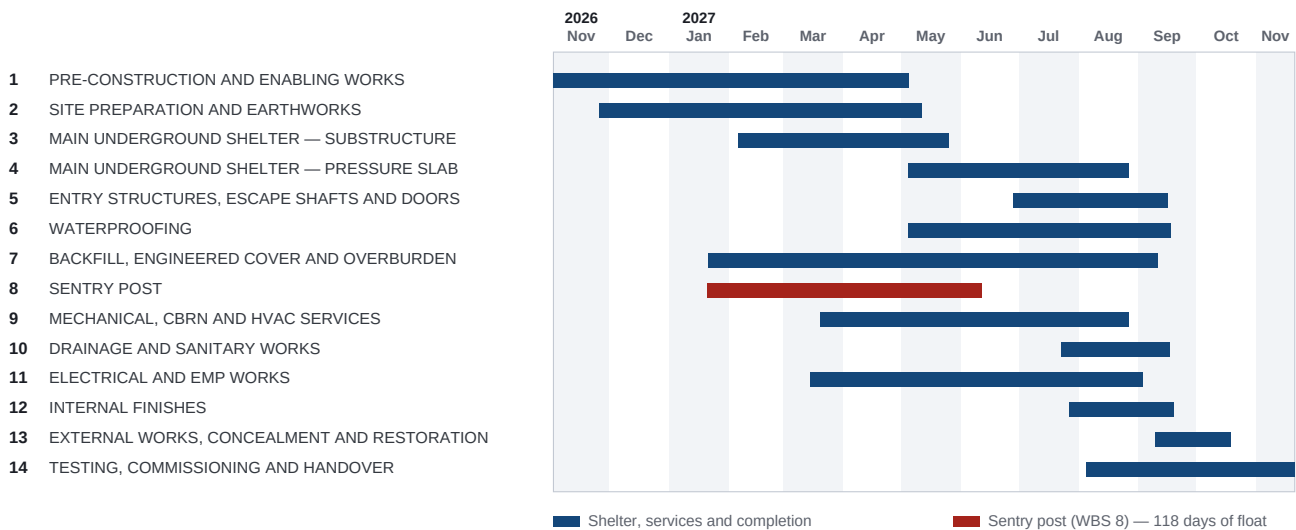


Figure 10.1 — Summary programme, the fourteen level-1 work packages.

10.1 Calendar

Six-day working week, Monday to Saturday; Sunday non-working. Five date-certain national holidays are non-working. Maharashtra festival holidays are NOT in the calendar because their dates move; the programme carries a stated contingency activity before handover to absorb them. Monsoon (June to September at Pune) is handled by productivity allowances inside the affected durations and by the sequencing rule that the deep excavation and the external tanking are kept out of the monsoon, not by making the calendar non-working.

The six-day week is read from the preliminary schedule rather than imposed: its stated 224 working days between 2 November 2026 and 26 July 2027 corresponds to a six-day week with a small number of holidays, and the same convention is kept.

10.2 Milestones

Ref	Milestone	Date	Float	Status
M-01	Site possession and commencement	07 Nov 2026	0 d	CRITICAL
M-02	Mobilisation complete	26 Nov 2026	8 d	
M-03	All long-lead protective items ordered	23 Feb 2027	230 d	
M-04	Excavation complete, formation approved	06 Feb 2027	0 d	CRITICAL
M-05	Mat foundation cast	15 Mar 2027	213 d	
M-06	Perimeter and protective walls complete	05 May 2027	170 d	
M-07	Pressure slab cast — key structural milestone	28 Jun 2027	26 d	
M-08	Buried envelope watertight — tanking complete	05 Aug 2027	91 d	
M-09	Overburden complete — structure buried	10 Sep 2027	0 d	CRITICAL
M-10	Sentry post frame complete	20 Apr 2027	182 d	
M-11	Sentry post brick masonry complete	28 Apr 2027	175 d	
M-12	Sentry post complete	11 Jun 2027	109 d	
M-13	CBRN plant installed	26 Aug 2027	23 d	
M-14	Drainage installed	16 Sep 2027	13 d	
M-15	Electrical installation complete	02 Sep 2027	22 d	
M-16	Internal finishes complete	18 Sep 2027	53 d	
M-17	External works and concealment complete	18 Oct 2027	0 d	CRITICAL

Ref	Milestone	Date	Float	Status
M-18	PRACTICAL COMPLETION AND HANDOVER	20 Nov 2027	0 d	CRITICAL

Table 10.1 — Milestone register.

10.3 Critical path and float

75 activities carry zero total float. The controlling chain runs: site possession, the confirmatory site investigation and monsoon groundwater monitoring — which must close the assumed water table before the excavation support can be designed — then rock excavation, formation approval, the external tanking, the mat, the perimeter and protective walls, the pressure slab with its fourteen-day prop period, the roof membrane, the six-layer engineered cover including the burster slab, the external works and concealment, and finally snagging, rectification and handover.

Total float band	Activities	Interpretation
TF = 0	75	Critical — any slip is a project slip
TF 1–10	36	Near critical — flagged in every monthly report
TF 11–30	57	Manageable
TF 31–90	46	Comfortable
TF > 90	65	Large — procurement and the sentry post

Table 10.2 — Float distribution.

10.4 Programme file

The editable master programme is issued as `Underground_Shelter_Final_Works_Programme.xml`, written in the Microsoft Project Data Interchange schema — Microsoft Project's own published XML format. Microsoft Project opens it directly, and *File* → *Save As* → *Project (*.mpp)* produces the binary file. The native `.mpp` format is undocumented and can only be written by Microsoft Project itself, so it is not imitated here. The interchange file carries the full breakdown structure, every activity, every logic link with its lag, the six-day calendar with its exceptions, 33 resources and 426 assignments, and it was verified by reading it back and comparing every date against the critical-path calculation. A programme drawing and a comma-separated task list accompany it.

11 Bill of quantities

The bill covers the entire project. Nothing in it is fabricated: every quantity is either taken directly from an existing project package or derived from confirmed geometry with the arithmetic shown. The full derivation — the source of every input and every line of working — accompanies this handout.

	Section	Items	Principal quantities
A	SITE PREPARATION AND EARTHWORKS	11	1 338 m ³ bulk excavation · 994 m ³ in rock · 290 m ³ backfill
B	MAIN UNDERGROUND SHELTER — CONCRETE	14	388.7 m ³ M35 · 17.3 m ³ M15 blinding
C	REINFORCEMENT — ALL WORKS	13	77.33 t ordered · 70.46 t shelter · 1.97 t sentry · 1.22 t burster
D	FORMWORK AND FALSEWORK	2	1 058 m ² formwork · 104 m ² falsework deck at 3.200 m
E	ENGINEERED COVER, OVERBURDEN AND CONCEALMENT	8	205.8 m ³ in six layers · 20.6 m ³ M30 burster slab
F	WATERPROOFING	9	532 m ² tanking · 110 m waterstop
G	SENTRY POST — CONCRETE AND FRAME	10	20.95 m ³ M30 · 4 No. footings F1
H	SENTRY POST — BRICK MASONRY AND FINISHES (design change SP-B1)	13	12.20 m³ brick masonry · 64.19 m² face · about 6 400 bricks
I	ELECTRICAL AND EMP INSTALLATION	7	All quantities to be verified — no electrical design exists
J	PRINCIPAL MATERIALS — PROCUREMENT VOLUMES	3	194 t cement · 829 t aggregate

Table 11.1 — Bill summary.

11.1 Basis of measurement

- Measurement follows the IS 1200 series — Part 1 earthwork, Part 2 concrete, Part 3 brickwork, Part 5 formwork, Part 12 plastering.
- Concrete volumes for the shelter and entry structures are taken verbatim from the existing reinforcement package. Six of its eight lines were independently re-derived here and reproduce exactly.
- Reinforcement for the shelter is taken verbatim from the approved bar bending schedule: 70.46 t across 92 bar marks, with that package's own declared convention — cut length as the sum of scheduled legs with no bend deduction, laps at 50 ϕ , 12 m stock bar.
- The sentry post had never been quantified anywhere in the project. Its concrete and reinforcement are derived here from the confirmed arrangement in the master reinforcement register, for tender and estimating only.
- Rates are not included. Rates come from the MES Standard Schedule of Rates, and no item number from it is quoted because none could be verified. To be verified against the applicable MES SOR/SSR/specification edition.

11.2 The sentry post brick masonry

	Ground storey	First storey	Total
Wall run between columns	15.200 m	15.200 m	—
Panel height	2.600 m	2.600 m	—
Gross face area	39.52 m ²	39.52 m ²	79.04 m ²
Openings deducted	3.33 m ²	11.52 m ²	14.85 m ²
Net masonry face	36.19 m²	28.00 m²	64.19 m²
Volume at 190 thk	6.88 m³	5.32 m³	12.20 m³
Bricks including 5 % breakage	—	—	about 6 400 No.
Cement for CM 1:6	—	—	14.5 bags
Sand	—	—	3.0 m ³

Table 11.2 — Brick masonry measurement.

11.3 What cannot be quantified

Where a quantity genuinely cannot be determined from the project record, the item reads *to be verified from final measurement* and the reason is stated. It is not estimated and it is not left out.

Item	Why
Berm volume, regrading, access route, hardstanding	No site plan or ground model exists
Concealment beyond the topsoil layer	No concealment specification exists
Every electrical quantity	No electrical design package exists — the scope is confirmed, the design is not
Blast door leaves, frames and anchorage	Vendor items, recorded in the project as not available
Blast valve sizes and sleeve details	Not in the structural design record
Sentry post lintel design	Does not exist — a consequence of SP-B1
Sentry post ground-floor slab	Not designed; the member schedule covers the first floor and roof only
Finishes rates	The finishes schedule specifies performance and leaves the product deliberately open, so areas can be measured but rates cannot
Window and vision-panel heights	Not stated on any drawing

Table 11.3 — Items declared as not determinable.

12 Resource management

The resource plan lists the resources that actually do the work and nothing else. Every resource below is assigned to at least one activity in the programme; a plan padded with resources that never appear on the programme is not a plan.

12.1 Staff

Resource	Composition	Peak	What drives the peak	Acts
Project Manager	1 No.	1	Overall responsibility; MES/Service liaison; procurement of protective items	18
Planning Engineer	1 No.	1	Programme, progress measurement, look-ahead, as-built records	2
Site Engineer (Civil)	2 No.	2	Excavation, RC works, backfill, sentry post	15
Site Engineer (Services)	1 No.	1	HVAC, CBRN, drainage, electrical coordination	3
Surveyor	1 No. + chainman	1	Setting out to the master A.4.1 grid; level control; as-built survey	9
QA/QC Engineer	1 No.	1	Inspection and test plan, hold points, records	32
Safety Officer	1 No.	1	Deep excavation, confined space, work at height, lifting	2
Storekeeper	1 No.	1	Material receipt, stacking to IS 4082, issue	1
Structural Consultant	visiting	1	Attends the reinforcement hold points — carried from the R0 schedule	6

12.2 Labour

Resource	Composition	Peak	What drives the peak	Acts
Bar bender gang	1 gang = 1 fitter + 3 helpers	4	Peak on the pressure slab: 20.5 t of T25 at 150 both faces both ways	44
Carpenter gang	1 gang = 1 carpenter + 2 helpers	4	Peak on the pressure slab falsework, 104 m2 of deck at 3.200 m	37
Concrete gang	1 gang = 1 mason + 6 helpers	2	Continuous pours: mat 85 m3, pressure slab 112 m3	38
Mason gang	1 gang = 1 mason + 2 helpers	3	SENTRY POST BRICK MASONRY; screeds; floor finishes; chambers	15
Plasterer gang	1 gang = 1 plasterer + 1 helper	2	Sentry post internal and external plaster	3
Painter gang	1 gang = 1 painter + 1 helper	2	Sealed washable coatings internally; sentry post painting	4
Waterproofing gang	specialist, 1 gang = 4 No.	4	Tanking is a specialist trade — 532 m2 of continuous membrane	10
Electrician gang	1 gang = 1 electrician + 1 helper	3	Conduits, cabling, earthing, EMP bonding	22
Plumber gang	1 gang = 1 plumber + 1 helper	2	Drainage, sumps, rising mains	16
HVAC fitter	1 gang = 1 fitter + 1 helper	2	Ductwork, blast valves, CBRN plant	11
Unskilled gang	general labour	20	Curing, handling, backfill, cleaning	48
Specialist subcontractor	as required	1	EMP enclosure, blast doors, CBRN commissioning, gas-tightness test	8

12.3 Plant and equipment

Resource	Composition	Peak	What drives the peak	Acts
Excavator	0.9 m3 tracked	2	Bulk excavation, backfill, cover	17
Rock breaker	hydraulic, excavator mounted	2	994 m3 of basalt; NO BLASTING permitted near the sentry post	5
Tipper	10 t	4	Spoil haulage; peak during rock excavation	5
Dozer/grader	1 No.	1	Berm, regrading, access	4
Vibratory roller	1 No.	1	Backfill and cover compaction to 95 % MDD	5
Plate compactor	2 No.	2	Confined backfill in 250 layers	5
Concrete pump	boom pump	1	112 m3 continuous pour to the pressure slab	8
Transit mixer	6 m3	4	Peak on the pressure slab pour	4

Resource	Composition	Peak	What drives the peak	Acts
Crane	20 t mobile	1	Blast doors, filter trains, EMP enclosure, spiral stair	6
Dewatering pumps	duty + standby	2	Continuous through the substructure; GWT (-)2.000 is ASSUMED	2
Scaffolding	tube and fitting	1	Slab falsework, masonry, plaster; IS 3696 / IS 4014	8
Welding set	2 No.	2	EMP straps and blast door frames welded to the reinforcement cage	13

12.4 Resources for the sentry post masonry

The masonry gang is three masons with helpers. At roughly 2.4 m² of finished one-brick wall per mason-day — reduced from the flat-wall rate because both storeys are short panels between columns with a high proportion of opening perimeter and cut brick — three masons clear the ground storey in five days and the first storey in four. The first storey has less masonry but more openings, more cutting and more scaffold, which is why the saving is one day and not two.

All output rates in this package are assumed. The quantities they are applied to are confirmed or derived and are traceable; the rates are not, and they are the first thing that should be re-tested when a contractor is appointed.

12.5 Peak resource periods

Period	Peak demand	Why
Dec 2026 – Jan 2027	2 excavators, 2 rock breakers, 4 tippers	994 m ³ of basalt removed by breaker in 20 working days
Mar – May 2027	4 bar-bender gangs	24.7 t of wall reinforcement at 150 spacing in both curtains
May – Jun 2027	4 bar-bender and 4 carpenter gangs	Pressure slab — 20.5 t of T25 and 104 m ² of deck at 3.200 m
25–26 Jun 2027	2 concrete gangs, pump, 4 transit mixers	The 112 m ³ continuous pressure slab pour
Mar – Apr 2027	3 mason gangs	Sentry post brick masonry, both storeys
Aug – Sep 2027	Excavator, roller, plate compactors	The six-layer engineered cover, 206 m ³
Sep – Oct 2027	Specialist subcontractors	Shielding survey, blast door tests, gas-tightness, CBRN commissioning

Table 12.1 — Peak resource periods.

13 Procurement

Procurement is programmed, not assumed. Every package is tied to an activity in the master programme and every package is required before the activity that installs it, with the dates computed from the programme rather than typed in.

WHY PROCUREMENT IS CONSTRUCTION LOGIC HERE

The governing fact of this project's procurement is that **the blast door cast-in frames must be welded into the W6 and W7 reinforcement cages before those walls are poured**. A seventy-five-day vendor lead time therefore sits directly upstream of the wall concrete, which is why the blast door enquiry is the first procurement activity on the programme and why the frames are ordered as a separate, earlier delivery than the door leaves.

Package	Lead	Ordered	Delivered	Needed	Float
Blast door cast-in frames	75 d	12-12-26	11-03-27	01-04-27	17 d
Blast door leaves	120 d	12-12-26	04-05-27	20-07-27	65 d
CBRN filter trains	100 d	12-12-26	09-04-27	04-08-27	98 d
Blast valves and sleeves	100 d	12-12-26	09-04-27	17-06-27	57 d
Shielded enclosure	80 d	18-12-26	23-03-27	23-07-27	103 d
Generator 15 kVA	60 d	19-11-26	29-01-27	24-07-27	149 d
Armoured vision panels	80 d	19-11-26	22-02-27	18-05-27	71 d
Submersible pumps	45 d	19-11-26	11-01-27	03-08-27	172 d
Reinforcement, 77.3 t	30 d	19-11-26	23-12-26	19-02-27	47 d
Waterproofing system	30 d	19-11-26	23-12-26	10-02-27	39 d
Modular bricks to IS 1077	20 d	19-11-26	11-12-26	15-03-27	77 d

Table 13.1 — Long-lead items. All lead times are assumed; replacing them with quoted times at tender is the single most valuable thing that can be done to de-risk this programme.

13.1 Brick masonry procurement

The whole brick requirement is about 6 400 bricks — a single approved batch. That is an advantage worth protecting. The source is approved once, against IS 1077 class designation 10, with samples drawn to IS 5454 and tested to IS 3495 for compressive strength, water absorption, efflorescence and warpage *before* the order is placed, and it is not changed without retesting. Changing brick source part way through a 64 m² job means retesting for a quantity too small to justify it.

13.2 Procurement risk

Four packages have no recovery available once they are late: the blast doors, the CBRN filter trains, the blast valves and the shielded enclosure. They are managed by monitoring the vendor — monthly manufacturing progress reports — rather than by planning to recover, because there is no recovery to plan.

One package cannot be procured at all yet. No electrical design package exists, so no electrical enquiry can be issued. The date by which the design is needed is computed from the programme and issued to the designer at mobilisation rather than requested when the activity falls due.

14 Quality assurance and quality control

The quality plan is built around hold points. A hold point is an activity in its own right in the programme — it takes time, it requires the Engineer's attendance, and work does not proceed past it. There are 16 of them, placed at the moments where an error becomes permanent.

Hold	Activity	Description	Date
H-11	A8020	founding stratum inspection — in-situ basalt at (-)2.000	29-01-27
H-01	A2085	formation inspection and approval by the Engineer	04-02-27
H-12	A8035	footing reinforcement and starter inspection	04-02-27
H-02	A3030	tanking integrity inspection before covering	18-02-27
H-03	A3070	mat reinforcement, cover and cast-in inspection	11-03-27
H-13	A8080	first floor reinforcement inspection — IS 13920 detailing	12-03-27
H-15	A8145	masonry line, level, plumb and joint thickness — ground storey	31-03-27
H-14	A8105	sentry post roof reinforcement inspection	06-04-27
H-04	A3120	wall reinforcement, frames and cover inspection — lift 1	08-04-27
H-05	A3140	wall reinforcement and cover inspection — lift 2	22-04-27
H-16	A8160	masonry inspection — first storey	27-04-27
H-09 — the last chance to see it	A6025	vertical tanking inspection before backfilling	24-05-27
H-06	A4070	pressure slab pre-pour inspection — reinforcement, cover, cast-ins	23-06-27
H-10	A6040	roof membrane integrity test before covering	04-08-27
H-07	A5030	headhouse wall inspection	07-08-27
H-08	A5055	headhouse roof inspection	26-08-27

Table 14.1 — Hold points in the programme.

14.1 The five checks that matter most

- **The formation.** The hazard is not the basalt, it is the red-bole and vesicular seams at the flow contacts — a single seam under the mat produces the differential-support case that sizes the mat. Every seam is over-excavated and replaced with M15, and the approval carries a seam map.
- **The tanking before it is covered.** The external tank is continuous under the mat, up the walls and over the roof. Once backfilled it can never be seen again, and the structure sits with its roof 2 m below grade and a design water table at roof level.
- **Reinforcement at 150 spacing in both curtains.** This is an electromagnetic pulse requirement and is stricter than IS 456. It is not a strength requirement, which means an engineer looking only at strength will see no reason not to relax it when the cage gets congested. It must not be relaxed.
- **Formwork striking.** Props to a slab spanning over 4.5 m stay for 14 days; this slab spans 5.000 m clear. Striking requires a signed permit and the fourteen days are in the baseline programme.
- **Sentry post masonry line, level and plumb.** Checked at the end of each storey, before plaster. Plumb over the 2.600 m storey height and joint thickness decide whether the plaster can be held to its stated thickness — and once plaster is on, neither can be corrected.

14.2 Quality control of the brick masonry

The masonry brings a set of checks the reinforced concrete panels did not need. They are grouped so that they can be issued as a single inspection sheet.

Check	Criterion	Reference
Brick source approval	Class 10; dimensional tolerance on 190 × 90 × 90	IS 1077
Compressive strength	Not less than 10 N/mm ² average	IS 3495 (Part 1)

Check	Criterion	Reference
Water absorption, efflorescence, warpage	Within the IS 1077 limits for the class	IS 3495 (Parts 2–4)
Sampling	Sample size and selection	IS 5454
Soaking	Thoroughly wet before laying, not surface-damp	IS 2212
Mortar	CM 1:6 by gauge box, used within 30 minutes, never re-tempered	IS 2250
Mortar sand	Grading and cleanliness	IS 2116
Bond and joints	English bond one brick thick; 10 mm joints fully filled bed, cross and vertical	IS 2212
Lift height	Not more than 1 m above adjacent work in a day	IS 2212
Level and plumb	Level every fourth course; plumb over the full 2.600 m storey height	IS 2212 — hold point
Frame interface	Top course wedged and packed only after initial shrinkage	IS 1905 — hold point
Wall ties	At the specified spacing	Detail does not yet exist
Lintels	Approved design, section, reinforcement, 200 bearing each end	IS 456 — design does not yet exist
Curing	Damp for at least seven days	IS 2212
Chases for conduit	Cut, not hammered; not before the masonry has cured	IS 2212
Plaster	Thickness, line, level, no hollowness on tapping, seven-day cure	IS 1661

Table 14.2 — Brick masonry quality control.

14.3 Non-conformance

A non-conformance is raised the moment it is identified, by anyone, and is closed only against evidence. Three categories on this project are not closable by repair and require the designer's ruling: reinforcement spacing relaxed beyond 150 mm inside the protective envelope; a construction joint formed without both waterstops and the bonding strap; and any penetration cut through the envelope after casting.

15 Safety management

Four features set the safety regime for this project: a 6.8 m deep excavation in rock, a buried structure that becomes a confined space, substantial work at height on the headhouse and the sentry post, and specialist lifting of heavy protective components.

Baseline arrangements are site induction before first entry; minimum personal protective equipment of helmet, safety footwear, high-visibility clothing and eye protection with task-specific additions; a first-aid post and trained first-aiders; a documented emergency plan covering rescue from height, rescue from confined space and evacuation of a flooded excavation; a weekly safety walk and a monthly safety committee; and permit-to-work for confined space entry, hot work and live electrical work.

Ref	Activity	Principal hazards	Controls	Reference
S-01	Deep excavation, 6.8 m	Collapse of the soil zone above rockhead; fall of persons and plant into the excavation; fall of material from the edge.	Battered or supported faces designed as temporary works before excavation starts; edge protection, barriers and signage installed as the excavation deepens; spoil and plant kept back from the edge; access by fixed ladders in protected bays; daily inspection by a competent person after rain.	IS 3764
S-02	Rock breaking	Flying fragments; noise; hand-arm and whole-body vibration; damage to adjacent structures.	Hydraulic breaking only, no blasting; exclusion zone around the rig; eye and hearing protection mandatory; vibration exposure monitored; a minimum 10 m separation from the sentry post founding level maintained.	IS 3764; IS 5121 for safety in erection — To be verified against the applicable MES SOR/SSR/specification edition.
S-03	Working in the excavation	Water accumulation; oxygen deficiency in a deep sump; entrapment.	Continuous dewatering with standby; the sump pit at (-)8.100 is treated as a CONFINED SPACE with permit, gas testing, ventilation, a standby person and rescue arrangements.	IS 3764; permit-to-work system
S-04	Reinforcement handling and fixing	Cuts and puncture from projecting bars; manual handling injuries; crush injuries during unloading.	Bar caps on all projecting verticals; mechanical unloading; gloves and safety footwear; bar-bending machines guarded; the reinforcement yard laid out so that bars are not carried over other trades.	IS 7969; IS 4082 for stacking
S-05	Formwork and falsework	Collapse of falsework under the 112 m³ pressure slab pour; falls from the deck at 3.200 m; falling components during striking.	Falsework designed, checked and inspected before every pour; a striking permit signed only against the IS 456 Table 11 periods; edge protection to the deck; exclusion zone below during striking.	IS 14687; IS 456 Cl. 11.3; IS 3696 (Part 1)
S-06	Concrete placing	Cement burns and dermatitis; eye injury; pump line whip; entanglement with vibrators; night work fatigue on continuous pours.	Gloves, eye protection and barrier cream; pump lines restrained and pressure-tested; a briefed pour team with defined rest breaks; lighting designed for the continuous pours.	IS 7969
S-07	Work at height — headhouse, sentry post, scaffolding	Falls from the scaffold and from the slab edges; falling tools and materials.	Scaffold erected, altered and dismantled only by trained scaffolders and tagged before use; full guard-rail, mid-rail and toe-board; harness where edge protection cannot be provided; toe-boards and debris netting.	IS 3696 (Part 1) and (Part 2); IS 4014 (Part 2)
S-08	SENTRY POST — brick masonry	Falls from the masonry scaffold at first-storey level; collapse of a freshly built panel; manual handling of bricks and mortar; silica dust from cutting; eye injury from mortar splash and from chasing.	Independent tied scaffold with a proper working platform, never a trestle on a slab edge; no more than 1 m of lift above adjacent work in a day, and temporary propping of a free-standing panel; mechanical hoisting of bricks above ground level rather than throwing; wet cutting or on-tool extraction and RPE for dust; eye protection for laying, cutting and chasing; the top course wedged only from a proper platform.	IS 2212 for the work itself; IS 3696 (Part 1); IS 7969
S-09	Lifting operations	Load failure during the lift of the blast door leaves, the filter trains, the EMP enclosure and the spiral stair.	Lift plan for every significant lift; certified crane, operator and tackle; exclusion zone; no lifting over occupied areas; wind limit set for the crane.	IS 3696 (Part 1); To be verified against the applicable MES SOR/SSR/specification edition.

Ref	Activity	Principal hazards	Controls	Reference
S-10	Confined space — sump, escape shafts, ducts	Oxygen deficiency; toxic atmosphere; entrapment in a 1400 dia shaft.	Permit to work, gas testing before and during entry, forced ventilation, harness and retrieval line, standby person, and a rehearsed rescue plan. The 1400 dia escape shafts are entered only under permit.	Permit-to-work system; To be verified against the applicable MES SOR/SSR/specification edition.
S-11	Electrical works	Electric shock; arc flash; working live.	All work dead unless a live-working permit is issued; lock-off and tag; residual current protection on all site supplies; only competent electricians; the generator changeover proved before energisation.	IS 732; IS 5216 (Part 1); IS 3043
S-12	Plant movement and haulage	Struck by excavator, tipper or roller; reversing incidents; interaction with pedestrians on a constrained site.	Segregated pedestrian routes; banksman for all reversing; reversing alarms and cameras; a one-way haul route; speed limit; daily plant inspection.	IS 7969; To be verified against the applicable MES SOR/SSR/specification edition.
S-13	Waterproofing and coatings	Solvent vapour in the confined internal envelope; fire; skin and respiratory sensitisation.	Forced ventilation during application in the closed envelope; no hot work adjacent; appropriate RPE and gloves; material safety data sheets held on site for every product; a fire watch where hot-applied systems are used.	Product safety data; To be verified against the applicable MES SOR/SSR/specification edition.
S-14	Backfill and cover placement	Plant overturning on the berm slope; crushing between plant and the structure; damage to the tanking membrane by plant tracking.	Compaction plant kept off the membrane until the 100 protection screed is cured; slope gradients controlled; banksman during placement adjacent to the structure.	IS 7969
S-15	Site-wide	General site hazards; emergency in a deep excavation or a confined space.	Site induction for every person; PPE as a minimum standard — helmet, safety footwear, high-visibility clothing, eye protection; first-aid post and trained first-aiders; documented emergency plan with rescue from height and from confined space; emergency contact list and assembly point; weekly safety walk and monthly safety committee.	IS 13416 (Parts 1 to 5); CPWD Safety Code; To be verified against the applicable MES SOR/SSR/specification edition.

Table 15.1 — Safety register.

15.1 Masonry hazards on the sentry post

Masonry is often treated as a low-risk trade, so the hazards it introduces are set out separately.

- **Falls from the masonry scaffold.** The first-storey panels are built 3.2 m above the ground floor. An independent tied scaffold with a proper working platform, guard-rail, mid-rail and toe-board is required — never a trestle standing on a slab edge.
- **Collapse of a freshly built panel.** A 190 mm panel 2.6 m high and up to 4.3 m long is slender while its mortar is green. Lift height is limited to 1 m above adjacent work in a day, and free-standing panels are temporarily propped.
- **Manual handling.** About 6 400 bricks and 3.5 m³ of mortar are handled. Bricks are hoisted mechanically above ground level, never thrown.
- **Silica dust.** A high proportion of the masonry is short panels between columns with cut brick at every opening reveal. Wet cutting or on-tool extraction, with respiratory protection.
- **Eye injury.** From mortar splash while laying the top course overhead, and from chasing for conduit. Eye protection for laying, cutting and chasing.

15.2 The three events that would stop the job

A fall of ground into the excavation while people work at (–)6.800 — controlled by designed faces, edge protection installed as the excavation deepens, and daily inspection after rain. **A falsework collapse** during the 112 m³ continuous pour, with a concrete gang working beneath 104 m² of deck at 3.200 m — controlled by designed and checked falsework inspected before every pour. **A confined-space fatality** in the sump at (–)8.100 or in a 1 400 mm escape shaft — controlled by permit, gas testing, forced ventilation, harness and retrieval line, a standby person and a rehearsed rescue plan.

16 Risk management

Risks are scored for likelihood and impact. The response column states what is actually done about the risk in the programme, the procurement plan or the quality plan — not a statement of intent.

Ref	Category	Risk	L	I	Response
R-01	Ground	Groundwater is higher than the ASSUMED (-)2.000. Water is nearly two-thirds of the 15.41 kPa/m lateral gradient, so the walls, the flotation check and the dewatering all move.	M	H	Monsoon monitoring (A1080) is deliberately on the critical path and must close before the excavation support is designed. If the level is higher, the design is referred back before excavation, not after.
R-02	Ground	Red-hole or vesicular seams found under the mat. A single seam produces the differential-support case that sizes the mat.	H	H	Confirmatory boreholes before excavation; every seam exposed at formation is over-excavated and replaced with M15; formation approval is a hold point with a seam map. Provisional quantity carried for replacement concrete.
R-03	Ground	Rock excavation quantity exceeds the estimate because rockhead is shallower than the mean assumed.	M	M	The BOQ gives the quantity at both bounds — 944 to 1043 m ³ — so the range is priced, not discovered. Roughly one extra day on the excavation.
R-04	Procurement	Blast door FRAMES arrive late. The frames must be welded into the W6 and W7 cages before those walls are poured, so a late frame stops the wall and the whole downstream structure.	M	H	Ordered on day one with a 75-day lead time and 19 days of float; manufacturing progress reported monthly; the frame is a separate delivery from the leaf so it is not delayed by leaf testing. This is the single largest procurement risk on the job.
R-05	Procurement	CBRN filter trains, blast valves or the EMP enclosure arrive late.	M	M	All ordered at mobilisation; each carries over 100 days of float; the installation activities sit after the structure is complete so a slip is absorbed rather than propagated.
R-06	Design	Conflict C21 — the CBRN filter duty is stated as 2 x 250 m ³ /h in master A.3 and as 300 m ³ /h in nine places on sheet S-06, and 250 fails S-06's own 264 m ³ /h criterion. UNRESOLVED.	H	M	Must be ruled on BEFORE the filter trains are ordered — the enquiry activity is programmed before order placement for exactly this reason. Escalated to the designer at mobilisation.
R-07	Design	Conflict C16 — the roof/platform junction, 250 against 500. UNRESOLVED in the master and not resolved by this package.	M	M	Affects the covered entry stairwell roof, which carries float. Ruling required before A5105. Raised in the verification register.
R-08	Design	No electrical design package exists. The scope is confirmed but no circuit, cable, luminaire, distribution board or earth-electrode schedule does.	H	H	Every electrical quantity is declared 'to be verified from final measurement'. The electrical design must be issued before A11025; the programme allows for it but cannot absorb an indefinite delay. Raised at mobilisation as the largest single information gap.
R-09	Design	SENTRY POST BRICK MASONRY (SP-B1) — no lintel design exists, no wall tie detail exists, and the seismic weight of the frame changes when RC infill becomes brickwork.	H	M	Lintel and tie design requested before A8135; the seismic re-check is referred to the structural discipline. The change reduces the infill line load from 13.000 to about 9.9 kN/m, which points the existing V _b = 73.18 kN in the conservative direction — but that is a direction, not a verification.
R-10	Design	The Rev F panels are described as BALLISTIC infill. Brick masonry does not give the same ballistic protection.	H	M	Recorded as a consequence of the instructed change so that the decision is visible. If ballistic performance is required it must be reinstated by another means; this package does not assume it has been.
R-11	Design	No site plan or ground model exists. The berm volume, access route, hardstanding, several drainage runs and the soakaway positions cannot be determined.	H	M	All affected items carried as 'to be verified from final measurement'. A site plan is requested at mobilisation; the external works sit at the end of the programme, which buys time but not an unlimited amount.
R-12	Construction	The 112 m ³ continuous pressure slab pour cannot be completed without a cold joint — plant breakdown, supply interruption or weather.	M	H	Standby pump and standby transit mixers arranged; the pour is planned as a single shift with defined fallback joint positions agreed with the designer IN ADVANCE, complete with waterstop and EMP strap details, so that an unplanned stop does not become an unplanned detail.
R-13	Construction	Reinforcement congestion at 150 spacing both curtains prevents proper compaction, particularly at the haunches, trimmer bands and blast door jambs.	M	H	The 150 spacing is an EMP requirement and must NOT be relaxed on site. Mock-up of the most congested zone before the pour; mix designed for workability at 0.45 w/c; small-diameter vibrators and a rehearsed placing sequence.
R-14	Construction	Damage to the external tanking membrane during backfilling.	M	H	Protection board or screed applied and inspected before any backfill; a hold point at the vertical tanking specifically because it can never be seen again; compaction plant kept off the membrane until the protection screed has cured.

Ref	Category	Risk	L	I	Response
R-15	Construction	SENTRY POST masonry quality — plumb, joint filling and the interface with the frame.	M	M	Bricks soaked before laying; gauge boxes for mortar; hold point on line, level, plumb and joint thickness at each storey; the top course wedged only after initial shrinkage so the panel does not attract frame load.
R-16	Materials	Brick supply — availability of class 10 modular bricks to IS 1077 of consistent size and strength.	M	L	Source approved and tested BEFORE order (A1176 precedes A8130); the whole requirement is only about 6 400 bricks, so a single approved batch covers the job. Changing source mid-job requires retesting.
R-17	Materials	Cement or reinforcement supply interruption during a continuous pour period.	M	M	Reinforcement ordered at 77.3 t including wastage with mill certificates; cement stocked to cover the largest pour plus a margin; stacking to IS 4082 to prevent deterioration.
R-18	Weather	Monsoon — Pune, June to September. Deep excavation, tanking and cover placement are all weather sensitive.	H	M	Sequenced so that the deep excavation (Dec–Jan) and the external tanking are out of the monsoon; surface water cut-off drains installed before excavation; productivity allowances inside the affected durations rather than an artificial non-working calendar.
R-19	Programme	Festival holidays are not in the calendar because their dates move.	H	L	A 10-working-day contingency activity (A14098) sits before handover for exactly this purpose. It is stated, not hidden inside durations.
R-20	Programme	The 14-day prop period on the pressure slab is questioned on site and props are struck early.	L	H	IS 456 Table 11 is explicit for a slab spanning over 4.5 m; this one spans 5.000 m clear. Striking requires a signed permit. The 19 days are in the baseline and are not available for compression.
R-21	Quality	Percolation test fails — likely on basalt — and the soak pits cannot work. SK-01 is already 2.3 % short of its own stated requirement (conflict C19, OPEN).	H	M	The percolation test is programmed early (A1085) and is a hold point before the soak pits are built. If it fails, an alternative disposal route is a design decision, not a site one, and the programme flags it early enough to be taken.
R-22	Testing	EMP shielding effectiveness survey fails to reach 80 dB.	M	H	The rebar cage alone gives 0 dB at 1 GHz — the master says so. The answer is the welded steel Zone 2 room plus disciplined penetration protection and joint straps, all of which are inspected as they are built rather than tested for the first time at the end.
R-23	Testing	Gas-tightness and overpressure test fails.	M	H	Every penetration through the envelope is recorded and sealed as built; the blast door seals are proved before the envelope test; the test is programmed with rectification time and float ahead of handover.
R-24	Interface	Damage to the completed sentry post by main-works traffic during the remaining eight months of the programme.	M	L	The sentry post carries 118 days of float and is deliberately started only after the rock breaking finishes, which shortens its exposure. Its surround and apron are built at the end with the external works.
R-25	Access	Constrained site access on an operational military installation; restricted working hours or short-notice closures.	M	M	Access control established first; deliveries booked; the programme uses a six-day week which leaves Sunday available for recovery without overtime premium on the critical path.

Table 16.1 — Risk register.

16.1 The four risks that carry the project

- **Blast door frames arrive late (R-04).** The frames must be welded into the W6 and W7 cages before those walls are poured. A late frame stops the wall, and the wall stops the pressure slab, the cover and everything after it.
- **No electrical design package exists (R-08).** The scope is confirmed but no circuit, cable, luminaire, distribution board or earth-electrode schedule does. This is the largest single information gap on the project.
- **Red-bolt seams under the mat (R-02).** The project record is explicit that the hazard is the flow contacts, not the basalt, and that a single seam produces the differential-support case that sizes the mat.
- **Groundwater higher than assumed (R-01).** Water is nearly two-thirds of the 15.41 kPa/m lateral gradient. Monsoon monitoring is deliberately on the critical path so the answer arrives before the excavation support is designed, not after the walls are built.

16.2 Risks arising from the design change

Substituting brick masonry for the reinforced concrete infill panels creates four risks that did not previously exist: there is no lintel design, no wall tie detail and a changed seismic weight (R-09); the ballistic function the panels were named for is lost (R-10); masonry workmanship becomes a quality risk on a structure that previously contained no masonry at all (R-15); and brick supply consistency matters for a quantity too small to justify retesting a second source (R-16).

The seismic point is worth stating precisely because it is easy to get wrong. The project uses a response reduction factor of 3.0 *specifically because* the infill is not separated from the frame, and designs to a base shear of 73.18 kN. Replacing 200 mm of concrete at 25 kN/m³ with 190 mm of brickwork at about 20 kN/m³ reduces the infill line load from 13.000 kN/m to roughly 9.9 kN/m, which reduces the seismic weight and therefore points the existing design in the

conservative direction. That is a direction, not a verification. The structural discipline must re-run the check; this package does not, and does not claim to.

17 Progress monitoring

The baseline is the WM1 programme: 279 activities, 326 working days, 2 November 2026 to 20 November 2027, with 18 milestones and 75 activities at zero total float.

The baseline is frozen at the start of construction. It is re-baselined only on a formal instruction — a change of scope, an extension of time or a design change — and every re-baseline keeps the previous version. A programme that is quietly re-drawn each month measures nothing.

Progress is reported against the baseline, never against the last revision of the programme.

17.1 Measurement

Physical progress is measured by QUANTITY, not by opinion. Each activity is linked to a BOQ item, so 'wall reinforcement 60 % complete' means 14.8 of 24.7 tonnes fixed and can be verified from the measurement book.

Earned value is computed on quantity: earned = quantity placed x BOQ rate. Schedule performance index and cost performance index are reported monthly for the project and for each of the three reporting streams.

Milestones are binary. A milestone is either achieved on its date or it is not; there is no percentage against a milestone.

17.2 Three reporting streams, one project

Progress is tracked separately for three streams and consolidated for the project. The streams are chosen so that each has a different trade, a different work face and a different risk profile; reporting them together would hide exactly the divergence that matters.

Stream	Scope	How it is measured	Baseline
A — Underground shelter	WBS 2 to 7 and 12. Excavation, substructure, pressure slab, entry structures, waterproofing, cover and internal finishes.	Measured on m ³ of excavation and concrete, tonnes of reinforcement, m ² of formwork and tanking, and m ³ of cover.	26-11-26 – 18-09-27
B — Sentry post	WBS 8. Frame, BRICK MASONRY WALLS, lintels, spiral stair, finishes and electrical.	Measured on m ³ of concrete, m ³ and m ² of brick masonry, and m ² of plaster and painting. Reported separately because it is an independent structure with 118 days of float.	21-01-27 – 11-06-27
C — Services and external works	WBS 9, 10, 11 and 13. HVAC and CBRN, drainage, electrical and EMP, external works and concealment.	Measured on installed lengths and items. Several quantities are 'to be verified from final measurement', so this stream is reported on activity completion until the design gaps close.	15-03-27 – 18-10-27

Table 17.1 — Reporting streams.

17.3 Reporting cycle

Frequency	Content
Daily	Site diary; labour and plant returns; concrete pour cards; weather; any stoppage and its cause.
Weekly	Progress against the three-week look-ahead; quantities placed; hold points cleared and outstanding; a short-interval programme for the coming week issued every Saturday.
Fortnightly	Rolling three-week look-ahead reissued, showing the activities starting and finishing, the resources needed, the materials required on site and the inspections due.
Monthly	Updated programme with actual start, actual finish and remaining duration on every activity; recalculated critical path; float erosion report; earned value for each stream and for the project; milestone status; a delay register with cause, effect and recovery proposal; the top ten risks reviewed.

Table 17.2 — Reporting cycle.

17.4 Delay tracking and corrective action

Every delay is recorded when it happens, with its cause, the activities affected and its effect on the critical path — not reconstructed at the end of the job.

Float erosion is tracked explicitly. An activity is flagged when its total float falls below ten working days, whether or not it is critical, because a chain that is quietly consuming float is the warning that arrives before a delay does.

Recovery is proposed in the same report as the delay: resequencing first, then additional resource, then additional shifts. Compressing the 14-day prop period on the pressure slab is not an available recovery measure and must not be offered as one.

Sentry post float — 118 days — is the natural place to absorb labour redeployment when the main works are held up, and the natural place to take labour from when the main works need to catch up.

The long-lead procurement chain has no recovery available once a delivery is late. It is therefore managed by monitoring the vendor, not by planning to recover: monthly manufacturing progress reports for the blast doors, filter trains, blast valves and the EMP enclosure.

Where an information gap is the cause — the electrical design, the site plan, the lintel design, conflicts C16 and C21 — the corrective action is escalation to the designer with a stated date by which the information is needed. Those dates are computed from the programme and issued at mobilisation, not requested when the activity is due to start.

18 Inspection and testing

The inspection and test plan runs from setting out to integrated commissioning. Inspection types are hold — work stops until released; witness — the Engineer is invited and work may proceed if attendance is waived; test — a measurement against a stated acceptance criterion; and surveillance — ongoing observation.

Ref	Work	Requirement	Reference	Type
Q-01	Site setting out	Position, orientation and level against the master A.4.1 grid; independent check. Sentry post at least 10 m clear of the excavation.	master A.4.1, A.2	WITNESS
Q-02	Excavation	Line, level and stability of the faces; edge protection in place; dewatering operating.	IS 3764	SURVEILLANCE
Q-03	Formation	Formation level (-)6.800; loose rock removed; red-bole and vesicular seams identified, over-excavated and replaced with M15. Founding stratum confirmed as in-situ basalt. The same check applies to the sentry post footings at (-)2.000 (H-11).	master A.6	HOLD (H-01, H-11)
Q-04	Backfill and engineered fill	Layer thickness not exceeding 250; in-situ dry density at least 95 % MDD.	IS 2720 (Part 8) for MDD; IS 2720 (Part 28) sand replacement	TEST
Q-05	Engineered cover build-up	Six layers reproduced at the thicknesses and materials of master A.7.3, totalling 2000 and 40.65 kPa.	master A.7.3	HOLD
Q-06	Concrete mix	Trial mix and design for M35 (box) and M30 (sentry). Minimum cement 340 kg/m ³ and w/c not more than 0.45 for M35 very severe exposure. Integral crystalline waterproofing admixture.	IS 456 Table 5; IS 10262; IS 2645	HOLD
Q-07	Cement, aggregate and water	Cement to IS 269 or IS 1489 (Part 1) with test certificates; aggregate to IS 383 tested to IS 2386; water to IS 456 Cl. 5.4.	IS 269, IS 383, IS 2386, IS 456 Cl. 5.4	TEST
Q-08	Reinforcement	Fe500D to IS 1786 for the box; Fe500 for the sentry post. Mill certificates plus independent tensile and bend tests per lot.	IS 1786	TEST
Q-09	Reinforcement fixing — mat	Bar size, spacing 150 both curtains, laps 50 phi staggered so not more than 50 % are spliced at a section, cover 75 to the blinding and 50 to formed faces, chairs and spacers, wall starters with a 900 leg.	IS 456 Cl. 26.2, 26.4.2, Table 16; master A.5, F.1	HOLD (H-03)
Q-10	Reinforcement fixing — walls	As Q-09 plus blast door jamb bars 4-T20 each face, header cage, haunch diagonals, and the blast door frame set, aligned and WELDED TO THE CAGE.	master F.1	HOLD (H-04, H-05, H-07)
Q-11	Reinforcement fixing — pressure slab	T25 @ 150 EF EW; opening trimmers 6-T25 and 5-T25; 4-T25 diagonals at the re-entrant corners; local thickenings 900 to 1200; four-legged T12 links @ 250 in the end 1500; HW3 band 4-T25 top and bottom; all cast-in items.	master B.4, B.4.1, F.1	HOLD (H-06, H-08)
Q-12	Formwork and falsework	Line, level, plumb, camber, tightness and cleanliness; falsework designed and checked as temporary works.	IS 456 Cl. 11; IS 14687	HOLD
Q-13	Concrete placing	Slump on every load; placing and compaction; no cold joint in a continuous pour; temperature control on the 900 slab and 600 mat.	IS 456 Cl. 13; IS 1199	SURVEILLANCE
Q-14	Concrete strength	Cube sampling and testing; acceptance to IS 456 Cl. 15 and 16.	IS 516; IS 456 Cl. 15, 16	TEST
Q-15	Curing	Continuous moist curing: 14 days for the mat, the pressure slab and the headhouse roof; 7 days minimum elsewhere.	IS 456 Cl. 13.5	SURVEILLANCE
Q-16	Formwork striking	Vertical faces after 16–24 h; props to slabs spanning over 4.5 m after 14 days. The pressure slab spans 5.000 clear.	IS 456 Cl. 11.3 Table 11	HOLD

Ref	Work	Requirement	Reference	Type
Q-17	Construction joints	Surface prepared; reinforcement fully continuous; TWO waterstops and a welded EMP strap installed before the next pour.	IS 456 Cl. 13.4; master F.1	HOLD
Q-18	Hardened concrete — NDT	Ultrasonic pulse velocity and rebound hammer on a sampling basis, concentrating on the pressure slab and the protective walls.	IS 13311 (Part 1) and (Part 2)	TEST
Q-19	Tanking — substrate	Substrate sound, clean, dry and free of laitance; fillets formed at all internal angles.	IS 3067	WITNESS
Q-20	Tanking — horizontal and vertical	Continuity, laps, terminations, upstands and dressing to every penetration; protection layer in place before backfilling.	IS 3067; FN1 WP-01/WP-02	HOLD (H-02, H-09, H-10)
Q-21	Watertightness	Visual inspection of the whole internal envelope for damp and seepage after the first monsoon exposure or a flood test where practicable.	IS 6494 principles	WITNESS
Q-21A	SENTRY POST — RC footings and frame reinforcement	Bar size, spacing, cover 50 to footings / 40 to columns / 30 to beams and slabs; column starters 8-T16; IS 13920 detailing — T10 confining hoops with a cross-tie each way @ 85 over 500 from every joint face AND through the joint; slab corner torsion steel T8 @ 200 in four layers over 700 x 700 at all four corners.	IS 456; IS 13920 Cl. 7.4, 8.1, 8.2; IS 456 Annex D-1.8; master F.4	HOLD (H-12, H-13, H-14)
Q-22	SENTRY POST — brick source	Class designation 10 modular bricks; sampling to IS 5454; testing to IS 3495 for compressive strength, water absorption, efflorescence and warpage. Dimensional tolerance check on the 190 x 90 x 90 size.	IS 1077; IS 3495; IS 5454	HOLD
Q-23	SENTRY POST — masonry mortar	CM 1:6 by volume using gauge boxes; sand to IS 2116; used within 30 minutes; never re-tempered. Mortar cube tests on a sampling basis.	IS 2250; IS 2116	TEST
Q-24	SENTRY POST — masonry workmanship	English bond one brick thick; 10 mm joints fully filled bed, cross and vertical; bricks thoroughly soaked before laying; not more than 1 m of lift above adjacent work in a day.	IS 2212	SURVEILLANCE
Q-25	SENTRY POST — line, level, plumb and joints	Plumb over the 2.600 storey height; level every fourth course; joint thickness 10 mm nominal; panel dimensions against the 190 thickness inside the 200 structural zone.	IS 2212; IS 1905	HOLD (H-15, H-16)
Q-26	SENTRY POST — infill / frame interface	Top course wedged and packed only after the panel has taken its initial shrinkage; ties to the columns installed at the specified spacing.	IS 1905; SP-B1	HOLD
Q-27	SENTRY POST — lintels	Approved lintel design in place before the opening is built; section, reinforcement, bearing 200 minimum each end, and curing.	IS 456; design NOT YET ISSUED	HOLD
Q-28	SENTRY POST — plaster	Surface prepared and raked; thickness, true to line and level; cured seven days; no hollowness on tapping.	IS 1661	WITNESS
Q-29	SENTRY POST — flooring and finishes	Level, falls, finish and freedom from cracking.	IS 2571 for in-situ cement concrete flooring	WITNESS
Q-30	SENTRY POST — painting	Surface preparation, primer and coats; adhesion.	IS 2395 (Part 1)	WITNESS
Q-31	Drainage — pipes and chambers	Line, level, gradient, jointing; water or air test; rodding access at every change of direction.	IS 1742 principles; Drainage DR1	HOLD
Q-32	Sump and pumps	Duty and standby pump performance; auto-alternation; level switch settings at start +900, stop +300 and high alarm +1200 above the invert; hand pump PU-03 proved independently of power.	Drainage DR1	TEST
Q-33	Soak pits	PERCOLATION TEST IS MANDATORY and must precede construction; SK-01 is already 2.3 % short of its own stated requirement (conflict C19, OPEN).	IS 2470 (Part 2) Cl. 4	HOLD
Q-34	Septic tank	Capacity, baffle at two-thirds of the length, inlet and outlet tees, 50 cowl vent at least 2 m above grade; watertightness.	IS 2470 (Part 1) Cl. 6.2, 6.3, 6.5, 6.6, 6.9, Table 1	WITNESS
Q-35	Electrical installation	Insulation resistance, continuity, polarity, protective device operation.	IS 732	TEST

Ref	Work	Requirement	Reference	Type
Q-36	Earthing	Earth electrode resistance against the 5 ohm target; bonding continuity to all metalwork.	IS 3043; IEEE 142	TEST
Q-37	EMP shielding	Shielding effectiveness survey, 80 dB over 10 kHz to 1 GHz.	IEEE Std 299; MIL-STD-188-125-1 \$5.4, \$5.5, \$5.7.2.1, \$5.7.4.1, \$5.7.6	TEST
Q-38	Blast doors	Alignment, seal compression, operation, and the vendor's rating documentation. The tested assembly is NOT overcoated.	Vendor specification; FN1 D-01	HOLD
Q-39	Gas-tightness and overpressure	Envelope leakage of bays 1 to 6 (67.8 m ² floor, 217.0 m ³ volume) and overpressure hold with the filter trains running.	FEMA 453 principles; project protective requirement	TEST
Q-40	CBRN filter trains	Airflow, filter seating, in-place leak test of the H14 HEPA stage, carbon bed condition, hand-crank operation of each fan.	EN 1822 for the HEPA classification; HVAC HV1	TEST
Q-41	Generator	Load test, changeover proving, exhaust and combustion air path, bunding.	HVAC HV1	TEST
Q-42	Integrated protective mode	Whole-system operation in protective mode: doors closed, blast valves set, filter trains running, overpressure maintained, drainage isolated.	This package	HOLD

Table 18.1 — Inspection and test plan.

18.1 The protective test sequence

The protective systems are tested in a fixed order because each depends on the one before it. The shielding effectiveness survey comes first, because it tests the enclosure and every penetration and there is no point proceeding if it fails. The blast door seals are proved next, because the envelope test depends on them. Then the gas-tightness and overpressure test of bays 1 to 6, then filter train commissioning, then the generator, then integrated protective-mode operation with the doors closed, the blast valves set, the filter trains running, overpressure maintained and the drainage isolated.

The shielding survey is also the test least likely to be recoverable if it fails, which is why the works it tests are inspected as they are built — bonding straps at every construction joint, door frames welded to the cage, penetration protection installed and recorded — rather than tested for the first time at the end.

19 External works

The external works complete the site: the berm against the above-ground structures, the regrading that sheds water away from a buried building, the surface water system, the access route, and the removal of everything temporary.

Act	Work	Days	Note
A13010	Berm forming and grading 1.5:1 to +0.900 against the headhouse and stairwell	8	volume NOT DETERMINABLE — no site plan exists
A13015	Site regrading — crown and falls 1:50 away from the structure	5	—
A13020	Surface water drainage, cut-off drains and outfalls	5	—
A13025	Access route, hardstanding and turning area	8	extent not defined in the project
A13030	Sentry post surround, apron and drainage	4	—
A13035	Turfing and planting over the cover — the concealment layer	6	the only concealment element the project confirms
A13040	Camouflage and concealment — spoil dressing, track discipline, screening	6	no concealment specification exists — scope to be confirmed
A13045	Removal of temporary works, site restoration and clearing	6	—

Table 19.1 — External works activities.

The berm is graded at 1.5:1 to +0.900 against the headhouse and the covered stairwell, and the site is crowned with 1:50 falls away from the structure. Surface water cut-off drains installed before excavation are retained and connected into the permanent system.

The extent of the external works cannot be quantified. No site plan or ground model exists in the project, so the berm volume, the access route and the hardstanding are programmed as activities with indicative durations and carried in the bill as items to be verified from final measurement. A site plan is requested at mobilisation; the external works sit at the end of the programme, which buys time but not an unlimited amount.

20 Electrical works

SCOPE CONFIRMED, DESIGN ABSENT

The electrical **scope** is confirmed by the project record: a 15 kVA generator in the grey zone, a shielded Zone 2 enclosure in the operations room, earthing to a 5 Ω target, bonding straps at every construction joint, and electromagnetic pulse protection at every service penetration to a requirement of 80 dB over 10 kHz to 1 GHz. **No electrical design package exists.** There is no circuit, cable, luminaire, distribution board or earth-electrode schedule anywhere in the project, so every electrical quantity in the bill reads *to be verified from final measurement* and no electrical enquiry can be issued until the design is produced.

The Works Management package still covers the electrical *works*, because the works are confirmed even though the quantities are not. They are programmed, resourced, given hold points and given tests.

Act	Work	Days	Note
A11015	Earth pits, earth electrodes and the buried earth ring	5	IS 3043; 5 ohm target
A11020	EMP straps and bonding at every construction joint	8	welded Cu/galv strap at every joint — a joint is an EMP discontinuity
A11025	Main cable route, service entry plate and EMP penetration protection	6	power PCI, waveguide below cutoff, fibre — MIL-STD-188-125-1
A11030	EMP Zone 2 shielded enclosure erection in bay 3 — SPECIALIST	12	the rebar cage gives 0 dB at 1 GHz; the welded steel room is the answer
A11035	Distribution board, sub-mains and final circuit wiring	12	IS 732
A11040	Lighting, small power and emergency lighting installation	10	—
A11045	Generator connection, changeover panel and protection	5	—
A11050	Earthing conductors, bonding and connection to the electrode system	6	—

Act	Work	Days	Note
A11055	Power to pumps, fans, filter trains and controls	6	—

Table 20.1 — Electrical and shielding activities.

The electromagnetic pulse works are as much structural as electrical, and that is why they appear early in the programme rather than at the end. Bonding straps are welded at every construction joint as the walls rise; the blast door frames are welded to the reinforcement cage before the walls are poured; the earth ring is laid before the mat is cast. The reinforcement cage on its own gives no useful attenuation at 1 GHz — the welded steel Zone 2 room is the whole answer, not a supplement to the cage — and the shielding effectiveness survey at handover tests all of it at once.

The sentry post has its own small electrical installation: conduits and boxes chased into the brick masonry after the masonry hold point, wiring and a distribution board, fittings and earthing, all tested before handover. Chases are cut, not hammered, and not before the masonry has cured — a detail that matters more in brickwork than it did in the reinforced concrete panels it replaces.

21 Camouflage and concealment works

Concealment is the point of a buried structure. A shelter whose position is obvious from the surface has lost part of its protection before the first weapon effect arrives.

The project confirms one concealment element: the 300 mm topsoil and turf layer at the top of the engineered cover, whose stated function is concealment, erosion control and shedding rain. It is programmed as the final layer of the cover, using the topsoil stripped at the start of the job and stockpiled for the purpose — which is why the stripping activity says *stockpile* and not *cart away*.

Beyond that layer, the project contains no concealment specification. Spoil dressing, screening, a planting regime and track discipline are all programmed as work, because they are the measures such a structure would normally need, but their extent and specification must be confirmed before they can be priced. The bill carries them as an item to be verified from final measurement rather than inventing a scope for them.

Two decisions elsewhere in the package serve concealment even though they are not labelled as concealment work. The engineered cover is graded and the site crowned with 1:50 falls so that the finished ground reads as ground rather than as a mound with drainage problems. And 994 m³ of rock is excavated while only about 51 m³ of crushed rubble is needed for the cover, so the surplus has to go somewhere — dressing it into the surrounding ground rather than leaving it as a spoil heap adjacent to a structure whose position is meant to be unremarkable is a concealment decision as much as a haulage one.

22 Completion and handover

Act	Work	Days	Start	Finish
A14075	Integrated systems commissioning — protective mode operation	6	29-09-27	06-10-27
A14080	Snagging inspection and defects list	5	18-10-27	22-10-27
A14085	Rectification of defects	10	23-10-27	03-11-27
A14090	As-built drawings, O&M; manuals and test certificates	12	07-10-27	20-10-27
A14095	Operator training — CBRN plant, pumps, generator, doors, escape shafts	5	21-10-27	26-10-27
A14098	Programme contingency — festival holidays and weather allowance	10	04-11-27	15-11-27
A14100	Final inspection and handover documentation	4	16-11-27	19-11-27
A14105	MILESTONE: PRACTICAL COMPLETION AND HANDOVER	—	20-11-27	20-11-27

Table 22.1 — Completion sequence.

Handover includes as-built drawings, operation and maintenance manuals, every test certificate, and operator training on the CBRN plant, the pumps, the generator, the blast doors and the escape shafts. The training matters more than usual here: the shelter has a duty pump, a standby pump *and* a hand pump precisely because it may have to work without power, and a fan with an electric drive *and* a hand crank for the same reason. Equipment provided as a third line of defence is only a third line of defence if the people who will use it have been shown how.

A ten-working-day contingency activity sits immediately before the final inspection. It exists because the calendar deliberately contains no fabricated festival-holiday dates: the movable holidays are absorbed there rather than hidden inside individual durations, where they would be impossible to audit.

23 Applicable codes and MES specifications

Every reference used in this package is a real, published document. No clause number, item number, schedule-of-rates reference, specification number or document title has been invented. Where an exact reference could not be verified from the material available, the entry says so in these words: *To be verified against the applicable MES SOR/SSR/specification edition.*

Structural design

Reference	Title	Used for
IS 456:2000	Plain and reinforced concrete — code of practice	All RC design and construction; Cl. 11.3 Table 11 formwork striking; Cl. 13.4 construction joints; Cl. 13.5 curing; Cl. 15 and 16 acceptance; Cl. 26.2 development and laps; Cl. 26.4.2 and Table 16 cover; Table 5 exposure, minimum cement and w/c.
IS 1786:2008	High strength deformed steel bars and wires for concrete reinforcement	Fe500D for the shelter, Fe500 for the sentry post.
IS 13920:2016	Ductile design and detailing of reinforced concrete structures subjected to seismic forces	Sentry post frame detailing, Zone III.
IS 1893 (Part 1):2016	Criteria for earthquake resistant design of structures — general provisions and buildings	Seismic design of the sentry post and the box.
IS 875 (Parts 1, 2, 3, 5)	Code of practice for design loads (other than earthquake)	Dead loads and unit weights; imposed loads; wind on the sentry post; load combinations.
IS 3370 (Parts 1 and 2):2021	Concrete structures for the storage of liquids	Crack width limit 0.2 mm and surface-zone steel.
IS 4991:1968	Criteria for blast resistant design of structures for explosions above ground	Blast loading rules and dynamic material strengths ONLY. Cl. 1.1 EXPLICITLY EXCLUDES NUCLEAR and that exclusion is quoted in every deliverable. Cl. 10.3.1.1 — NO dynamic increase on shear.
SP 34:1987	Handbook on concrete reinforcement and detailing	Shape codes; Cl. 5.5 opening-corner detailing.
SP 16:1980	Design aids for reinforced concrete	Column interaction charts, cross-checked against first-principles calculation.
BS 8666	Scheduling, dimensioning, bending and cutting of steel reinforcement for concrete	Bar shape codes and bend deductions in all bar bending schedules, used alongside SP 34.
IS 1904:1986	Design and construction of foundations in soils — general requirements	Presumptive bearing capacity, Table 1, hard rock 3240 kPa.
IS 12070:1987	Design and construction of shallow foundations on rocks	Settlement on sound basalt.

Materials

Reference	Title	Used for
IS 269:2015	Ordinary Portland Cement — specification	Cement for all concrete, mortar and plaster.
IS 1489 (Part 1):1991	Portland pozzolana cement — fly ash based	Alternative binder if approved.
IS 383:2016	Coarse and fine aggregate for concrete — specification	All concrete aggregate.
IS 2386 (Parts 1 to 8):1963	Methods of test for aggregates for concrete	Aggregate acceptance testing.
IS 2116:1980	Sand for masonry mortars — specification	SENTRY POST brick masonry mortar.
IS 1542:1992	Sand for plaster — specification	Sentry post internal and external plaster.
IS 2645:2003	Integral waterproofing compounds for cement mortar and concrete	The integral crystalline admixture required by master A.5.
IS 4031	Methods of physical tests for hydraulic cement	Cement acceptance testing. Multi-part standard — cite the relevant part for each test.

Construction

Reference	Title	Used for
IS 10262:2019	Concrete mix proportioning — guidelines	Trial mix design for M35 and M30.
IS 1199	Fresh concrete — methods of sampling, testing and analysis	Slump and sampling at the pour. Multi-part; confirm the current part and edition before citing a clause.

Reference	Title	Used for
IS 516	Hardened concrete — methods of test	Cube compressive strength. Confirm the current part and edition before citing a clause.
IS 13311 (Parts 1 and 2):1992	Non-destructive testing of concrete — ultrasonic pulse velocity; rebound hammer	NDT of the pressure slab and protective walls.
IS 14687:1999	Falsework for concrete structures — guidelines	Falsework to the pressure slab soffit at 3.200 m.
IS 9077:1979	Corrosion protection of steel reinforcement in RB and RCC construction	Reinforcement protection in a very severe exposure environment.
IS 2720 (Part 8):1983	Methods of test for soils — determination of water content–dry density relation using heavy compaction	MDD for the backfill and the engineered cover.
IS 2720 (Part 28):1974	Methods of test for soils — determination of dry density of soils in place by the sand replacement method	Field density testing of every compacted layer.

SENTRY POST masonry

Reference	Title	Used for
IS 1077:1992	Common burnt clay building bricks — specification	Class designation 10 modular bricks, 190 x 90 x 90, for the sentry post walls. DESIGN CHANGE SP-B1.
IS 2212:1991	Code of practice for brickwork	Laying, bond, joint filling, soaking, lift heights and curing.
IS 1905:1987	Code of practice for structural use of unreinforced masonry	Infill panel proportions, slenderness and the frame interface.
IS 2250:1981	Code of practice for preparation and use of masonry mortars	CM 1:6 preparation, gauge boxes, working time.
IS 3495 (Parts 1 to 4):1992	Methods of tests of burnt clay building bricks — compressive strength, water absorption, efflorescence, warpage	Brick acceptance testing before order and on delivery.
IS 5454:1978	Method for sampling of clay building bricks	Sample size and selection for the IS 3495 tests.
IS 1661:1972	Code of practice for application of cement and cement-lime plaster finishes	Sentry post internal and external plaster.
IS 2395 (Part 1):1994	Code of practice for painting concrete, masonry and plaster surfaces	Sentry post painting.
IS 2571:1970	Code of practice for laying in-situ cement concrete flooring	Sentry post flooring.

Waterproofing

Reference	Title	Used for
IS 3067:1988	Code of practice for general design details and preparatory work for damp-proofing and waterproofing of buildings	Substrate preparation, fillets, terminations for WP-01.
IS 6494:1988	Code of practice for waterproofing of underground water reservoirs and swimming pools	Principles applied to the buried envelope and the sump pit.

Drainage

Reference	Title	Used for
IS 2470 (Part 1):1985	Code of practice for installation of septic tanks — design criteria and construction	Septic tank ST-01; Cl. 6.2, 6.3, 6.5, 6.6, 6.9 and Table 1.
IS 2470 (Part 2):1985	Code of practice for installation of septic tanks — secondary treatment and disposal	Cl. 4 — the PERCOLATION TEST IS MANDATORY before the soak pits are built.

Electrical

Reference	Title	Used for
IS 732:2019	Code of practice for electrical wiring installations	Wiring, distribution and testing.
IS 3043:2018	Code of practice for earthing	Earthing installation and the 5 ohm target.
IEEE 142	Recommended practice for grounding of industrial and commercial power systems	Earthing target, cited alongside IS 3043.

Reference	Title	Used for
IS 694:2010	PVC insulated cables for working voltages up to and including 1100 V	Final circuit cabling.
IS 1554 (Part 1):1988	PVC insulated (heavy duty) electric cables for working voltages up to and including 1100 V	Sub-mains and generator cabling.
IS 3646 (Part 1):1992	Code of practice for interior illumination	Internal lighting levels.

Protective

Reference	Title	Used for
MIL-STD-188-125-1	High-altitude electromagnetic pulse protection for ground-based C4I facilities	EMP requirement: 80 dB over 10 kHz to 1 GHz; §5.4 access, §5.5 waveguide below cutoff, §5.7.2.1 power PCI, §5.7.4.1 fibre, §5.7.6 RF.
IEEE Std 299	Standard method for measuring the effectiveness of electromagnetic shielding enclosures	The shielding effectiveness survey at handover.
EN 1822	High efficiency air filters (EPA, HEPA and ULPA)	H14 classification of the CBRN filter train HEPA stage.
FEMA 453	Safe rooms and shelters	Shelter ventilation criteria and carbon adsorber performance.
UFC 3-340-02	Structures to resist the effects of accidental explosions	Cited AS US criteria: §4-27 opening trimmers, §4-30 direct shear.

Safety

Reference	Title	Used for
IS 3764:1992	Code of safety for excavation work	The 6.8 m deep excavation and its edge protection.
IS 3696 (Part 1):1987	Safety code for scaffolds and ladders — scaffolds	Masonry, plaster and slab-edge scaffolding.
IS 3696 (Part 2):1991	Safety code for scaffolds and ladders — ladders	Access into the excavation and to working platforms.
IS 4014 (Part 2):1967	Code of practice for steel tubular scaffolding — safety regulations	Erection, alteration and dismantling of tubular scaffold.
IS 7969:1975	Safety code for handling and storage of building materials	Reinforcement, brick, cement and aggregate handling.
IS 4082:1996	Recommendations on stacking and storage of construction materials and components at site	Site stacking discipline.
IS 13416 (Parts 1 to 5)	Recommendations for preventive measures against hazards at work places	General site safety, falling material, fire and excavation.
IS 5216 (Part 1):1982	Guide for safety procedures and practices in electrical work	Electrical safety and permit-to-work.

Life safety

Reference	Title	Used for
NBC 2016 Part 4	Fire and life safety	Stair geometry, 1100 guarding and means of escape.

Measurement

Reference	Title	Used for
IS 1200 (Part 1):1992	Methods of measurement of building and civil engineering works — earthwork	Measurement of excavation, filling and disposal.
IS 1200 (Part 2):1974	Methods of measurement — concrete works	Measurement of all concrete.
IS 1200 (Part 3):1976	Methods of measurement — brickwork	SENTRY POST brick masonry, measured at its actual 190 thickness.
IS 1200 (Part 5):1982	Methods of measurement — formwork	Formwork contact area.
IS 1200 (Part 12):1976	Methods of measurement — plastering and pointing	Sentry post plaster.
IS 1200 (Part 13):1994	Methods of measurement — whitewashing, colour washing, distempering and painting of buildings	Painting.

Government

Reference	Title	Used for
MES Standard Schedule of Rates (SSR), Part I — Rates and Part II — Specifications	The Military Engineer Services schedule governing measurement, specification and rates for Service works.	Measurement conventions, specification clauses and rates for every item of this BOQ. NO ITEM NUMBER IS QUOTED because none could be verified from the material available.
Defence Works Procedure (DWP)	The procedure governing the sanction, execution and acceptance of defence works.	Approvals, sanction stages, acceptance of works and handover to the user Service.
MES Regulations (MESR)	Standing regulations of the Military Engineer Services.	Contract administration, measurement, and the Engineer's authority.
CPWD Specifications 2019, Volumes 1 and 2	Central Public Works Department specifications for building works.	Workmanship clauses for earthwork, concrete, brickwork, plaster, flooring and painting where the MES SSR defers to them.
CPWD Works Manual	Central Public Works Department procedure for the execution of works.	Quality control, measurement books and inspection procedure.
Delhi Schedule of Rates (DSR)	CPWD schedule of rates used as a comparator where an MES SSR item is not available.	Rate comparison only. NO ITEM NUMBER IS QUOTED.

23.1 The position on MES and CPWD references

The Military Engineer Services Standard Schedule of Rates and the Defence Works Procedure are the documents that would actually govern a Service work of this kind. They exist, they are named correctly, and the work they would govern is identified. What is not given is any item number, clause number or rate, because none could be verified from the material available in this project — and putting a false authority on a document that an examiner or an executing engineer might rely on would be worse than leaving the gap visible. The same applies to the CPWD Specifications and the Delhi Schedule of Rates: the documents are real, the individual item numbers are not quoted.

24 References

Indian Standards, international standards and government specifications are listed in section 23 and in the accompanying codes register. The references below are the project documents that are the authority for everything this package measures and schedules.

Document	What it governs
Master project state file , Parts A, B, F and L	The single source of truth for geometry, materials, loading, structural design and reinforcement. Every quantity in this package traces back to it
Structural reinforcement package	Thirty A1 reinforcement drawings, 92 bar marks, 70.46 t of reinforcement and the concrete volumes used in the bill
Drainage package	Drainage design basis and the pipe, chamber, sump, pump and soakaway schedules
HVAC package	HVAC and CBRN design basis and the equipment, duct, filter and damper schedules
Schedule of finishes	Finish legend and rules for all thirteen spaces, specified as performance requirements
Revision F architectural drawings	The ten input drawings, including the sentry post ground floor, first floor and framing plans from which the wall runs and openings in section 8 were read
Structural analysis models	The underground plate model, the sentry post frame model and the entry stairwell frame model
Preliminary master construction schedule	The starting point for this programme; its start date and six-day calendar are retained

25 Conclusion

This package plans the construction and management of the complete project — the underground shelter, the entry structures, the escape shafts, the engineered cover, the sentry post and all of the services, external works and concealment works — from mobilisation to handover.

The logic runs in one direction and each step depends on the one before it. The project record establishes what is being built; the component register identifies the twenty-three components that actually exist; the work breakdown structure organises them into fourteen work packages; the construction methodology explains how each is built and why in that order; the bill of quantities measures them from confirmed geometry; the resource plan and the procurement plan provide what the quantities need; the programme sequences all of it into 279 activities across 326 working days; and the quality, safety, risk and progress systems control it while it happens.

Three things about the result are worth stating plainly. First, the programme is driven by a small number of facts that cannot be argued away: a fourteen-day prop period fixed by code, a seventy-five-day vendor lead time that sits upstream of a wall pour, and 994 m³ of rock that has to be broken rather than dug. Recognising those early is most of what distinguishes a programme that holds from one that does not.

Second, the package is honest about what it does not know. There is no electrical design, no site plan, no lintel design and no wall tie detail; several conflicts in the project record remain unresolved; and every output rate and vendor lead time is an assumption. All of it is listed, dated against the activity it blocks, and issued to the designer at mobilisation. A bill with visible gaps is more useful than one whose gaps have been filled with plausible numbers.

Third, the one design change carried by this package — the sentry post walls as brick masonry — has been followed through everywhere it reaches: into the methodology, the breakdown structure, the bill, the resource plan, the procurement plan, the programme, the quality plan, the safety register and the risk register. It has also been followed through into the places where it creates new obligations. Brick masonry needs lintels that the concrete panels did not; it needs ties that have not been detailed; it changes the seismic weight of the frame, conservatively but not verifiably; and it does not provide the ballistic protection that the panels it replaces were named for. Those consequences are recorded rather than absorbed, because a change whose consequences are invisible is the one that causes trouble on site.

Accompanying documents: work breakdown structure · bill of quantities and its full quantity derivation · resource plan · procurement plan · quality assurance and quality control plan · safety management and risk register · codes and references register · project component register · construction methodology · sentry post brick masonry methodology · progress monitoring and control · assumptions and verification register · master programme in Microsoft Project interchange format · programme drawing · comma-separated task list.